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REGULAR CONTRIBUTION

Machine learning methods for DDoS attack detection in software-defined networks

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Abstract

Distributed Denial of Service (DDoS) attacks remain among the most serious threats to Software-Defined Networks (SDNs), where centralised control makes timely detection essential. This paper presents a leakage-controlled comparison of six supervised classifiers, namely Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Decision Tree (DT), Random Forest (RF) and Extreme Gradient Boosting (XGBoost), and quantifies the effect of Mutual Information (MI) feature selection on detection performance and computational cost. Two open datasets are used: DDoS-SDN (104,345 flow records, binary) and APA-DDoS (151,200 packet records; benign, DDoS-ACK and DDoS-PSH-ACK). All preprocessing and feature selection are fitted on training data only, and models are evaluated on a stratified held-out test set using accuracy, precision, recall, specificity, F1-score, ROC-AUC and PR-AUC, supported by ten-fold cross-validation, McNemar tests with Holm correction and repeated timing. On DDoS-SDN with all 19 behavioural features, XGBoost reaches 100% test accuracy and RF 99.47%; the MI Top-12 subset retains near-full accuracy (XGBoost 99.99%, RF 99.40%), and the Top-5 subset raises KNN accuracy from 98.47% to 99.97% while reducing its latency from 0.76 to 0.007 ms per flow. A time-ordered split lowers accuracy by 6.7–17.8 percentage points for the non-linear classifiers, showing that random-split results overstate performance on later traffic. On APA-DDoS, a feature-level audit shows that the capture timestamp encodes the class label and that the TCP PSH flag defines the two attack classes. Once identifier and constant attributes are removed, the nine remaining features take only 65 unique combinations of values across all 151,200 records, and without the PSH flag no classifier can exceed 75.78% accuracy. APA-DDoS is therefore reported as a dataset audit rather than used to rank classifiers.

Keywords Distributed denial of service (DDoS) · Machine learning · Software-defined networking (SDN) · Traffic classification · Network security · Attack detection · Feature selection · Mutual information · Data leakage

1 Introduction

Cyberattacks have grown in both frequency and scale in recent years, and their increasing magnitude and complexity threaten the stability, security and resilience of the global digital infrastructure. Among these threats, DDoS attacks are particularly damaging and persistent. In a DDoS attack, a large number of compromised devices, typically organised into a botnet, are coordinated to overwhelm a victim such as a web server, an online service or a network [1]. The aim is to exhaust the target's resources so that its normal operation is disrupted and legitimate users can no longer reach the service. As more services move online and the number of connected devices continues to grow, DDoS attacks have become more frequent and more sophisticated, and they increasingly evade traditional defence mechanisms.

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Conventional security measures such as firewalls, rule-based systems and signature-based intrusion detection systems are often ineffective against modern DDoS attacks. They cope poorly with very high traffic volumes and with malicious traffic that is deliberately crafted to resemble legitimate activity. The growing use of encrypted traffic and the migration to IPv6 make anomaly detection still harder. Because these threats continue to evolve, there is a clear need for more effective, intelligent and responsive defences that can detect and react to attacks in a timely manner [2].

Over the past decade, machine learning has emerged as a promising response to these challenges. Unlike static rule-based systems, machine learning models learn from historical and current traffic and can recognise complex patterns and previously unseen attack variants. In DDoS detection, classifiers analyse statistical properties of network flows, such as packet and protocol distributions, flow duration and connection rates, to decide whether traffic is benign or malicious [3].

Supervised learning is a natural fit for DDoS detection. A model is trained and evaluated on a labelled dataset containing both normal and attack traffic, and the workflow typically comprises three stages, namely preprocessing, model training and performance evaluation, using metrics such as accuracy, precision, recall and F1-score [4]. The objective is an efficient and easily deployable classification method for detecting DDoS attacks in operational networks.

This study addresses several limitations of current DDoS detection techniques by developing machine learning models that can be integrated into modern intrusion detection systems and improve their ability to detect and respond to DDoS attacks [5]. The contribution is methodological and evaluative rather than a new classifier: established classifiers are compared under a leakage-controlled protocol, and the accuracy–cost trade-off of feature selection is quantified for SDN-based intrusion detection.

Figure 1 illustrates the mechanism of a DDoS attack. An attacker uses a botnet to send malicious requests to a large number of vulnerable DNS resolvers. These resolvers, acting as unwitting intermediaries, direct a high volume of traffic towards the target and exhaust its resources. Because the traffic originates from many distributed sources, the attack is difficult to trace and to mitigate, and it effectively denies legitimate users access to the target.

The main contributions of this work are as follows:

- We evaluate six supervised classifiers on the DDoS-SDN dataset in a single leakage-controlled pipeline in which imputation, encoding, Mutual Information feature selection and scaling are fitted on the training data only, and we report test-set accuracy, precision, recall, specificity, F1-score, ROC-AUC and PR-AUC together with the confusion matrices.
- We quantify the effect of MI-based Top-5 and Top-12 feature subsets on detection performance, training time, inference latency and throughput, showing that the Top-12 subset preserves near-full accuracy and that the Top-5 subset reduces KNN inference latency by two orders of magnitude.
- We assess the stability and robustness of the results with stratified ten-fold cross-validation, McNemar tests with Holm correction, a capacity check with unconstrained tree models, and three stricter splits (time-ordered, unseen source hosts and without topology attributes).
- We audit the APA-DDoS dataset at the feature level, identify identifier-based leakage and a construction-level definition of the attack classes, and quantify the redundancy of its feature space, showing why it should not be used to rank classifiers.

The remainder of the paper is organised as follows. Section 2 reviews related work, Sect. 3 describes the datasets, classifiers and evaluation methodology, Sect. 4 presents and

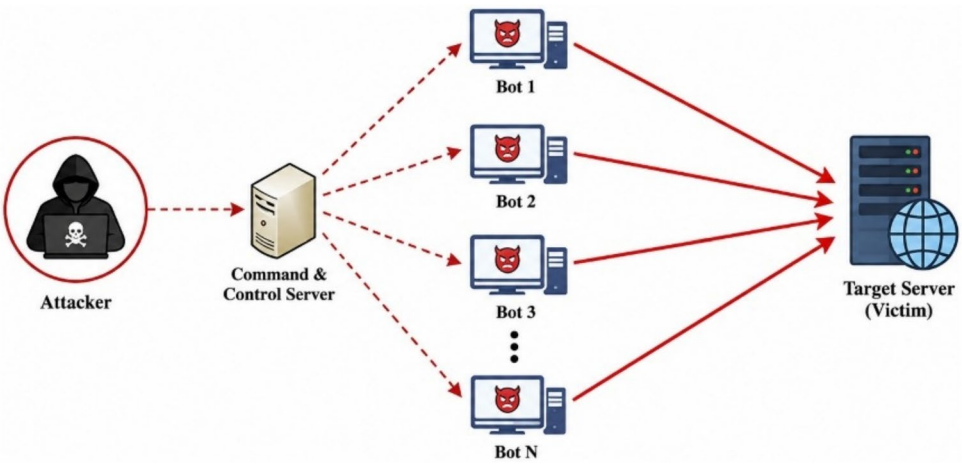


Fig. 1 Overview of a distributed denial of service attack

discusses the experimental results, Sect. 5 reports the statistical testing, computational cost, model-capacity check and comparison across datasets, Sect. 6 discusses the findings, and Sect. 7 concludes the paper and outlines future work.

2 Related work

In recent years, many studies have applied machine learning and deep learning to strengthen SDN security. The flexibility and programmability that make SDN attractive also expose it to a range of cyber threats, of which DDoS attacks are among the most serious and are the focus of this research. This section reviews representative machine learning and deep learning approaches to DDoS detection and mitigation.

Alashhab et al. [6] proposed an ensemble of a Decision Tree, Naive Bayes and a multi-layer perceptron for DDoS detection in SDN. Their system combines dynamic feature selection with real-time adaptation and, when evaluated on a Mininet/Ryu testbed, achieved an average detection rate of 99.2%. Hnamte et al. [7] developed a deep neural network for DDoS detection in SDN environments and reported detection rates of up to 100% on the CICIDS2018 and Kaggle datasets. Hussain et al. [8] introduced a hardware-assisted intrusion detection system (HRIDS) that couples rule-based logic with an ensemble of eight machine learning models operating on real-time industrial traffic, reporting a precision of 99.97% and a true positive rate of 99.96% for cyber-physical production systems.

Hassan et al. [9] combined entropy statistics with k-means clustering to detect DDoS attacks in SDN, using entropy variation to flag anomalies on the CICIDS 2017 and CIC-IDS 2019 datasets. Nisa et al. [10] proposed a two-phase authentication scheme based on a combination of SVM and KNN. Evaluated on CICDoS2017 in a Mininet-Ryu environment, it reduced CPU consumption, improved packet delivery and supported dynamic host revalidation. Kavitha et al. [11] addressed DDoS mitigation in cloud-based, SDN-controlled IoT networks and reported 99.99% accuracy against end device flooding on a multi-controller testbed. Ouhssini et al. [12] proposed DeepDefend, a CNN-LSTM-Transformer model for real-time DDoS monitoring in the cloud that uses a genetic algorithm for feature selection and achieved high detection accuracy on the CIDDS-001 dataset. Han et al. [13] applied feature selection to remove irrelevant traffic features for DDoS detection in 5G SDN and reported improved detection speed and accuracy on CICIDS 2017 and CICIDS 2018.

Paidipati et al. [14] proposed the MMEDRL-ADM framework, which combines deep reinforcement learning with African Buffalo and Grasshopper metaheuristics for feature selection and tuning, and reported high detection accuracy on benchmark datasets. Chaudhary et al. [15] developed a multi-phase detection scheme for fog-based consumer IoT networks in SDN that combines entropy detection, k-means clustering and Chebyshev thresholding, achieving 98.71% accuracy on the BoT-IoT dataset and reducing response time by 35% through a fog-based design. Santos-Neto et al. [16] proposed a hybrid machine learning and entropy model with dynamically adapted thresholds that exceeded 99% accuracy on DARPA and corporate datasets and converged faster than comparable RF and SVM baselines.

Kapourchali et al. [17] introduced P4httpGuard, a P4-based mechanism for detecting slow-rate DDoS attacks in SDN that reduced detection time as well as bandwidth and CPU consumption, demonstrating the feasibility of using programmable data planes against low-rate attacks. Abid et al. [18] proposed a CNN-LSTM hybrid model for DDoS detection in IoT-based SDN that captures both the spatial and the temporal structure of traffic and outperforms traditional machine-learning methods against evolving attack vectors.

John et al. [19] proposed the CBWSN_VSEMLA framework, which combines principal component analysis, fuzzy c-means clustering and ensemble learning for DoS detection in wireless sensor networks. On the WSN-DS dataset, a PCA-Random Forest configuration reached 99.9% accuracy with a favourable balance between detection performance and computational cost. Tonkal et al. [20] presented a neighbourhood component analysis-based detection model for SDN using classifiers such as a Decision Tree, SVM and an artificial neural network, with the Decision Tree reaching 100% accuracy on a public dataset of more than 100,000 flows.

Taken together, the surveyed studies reveal several research gaps that motivate this work. First, most existing approaches emphasise detection accuracy and give little attention to the associated computational cost. Second, there is limited analysis of how systematic feature reduction affects both detection performance and computational complexity across different feature subset sizes. Third, few studies compare a broad set of machine learning algorithms across more than one DDoS-SDN dataset to assess how dataset characteristics influence classifier behaviour. Finally, the generalisability and stability of the trained models are often validated with a single train-test split rather than cross-validation. In addition, few studies check whether a high reported accuracy is caused by identifier fields, duplicated records or attributes that define the class label, which can make a benchmark appear easier than the detection problem it represents.

Table 1 summarises representative state-of-the-art methods for DDoS detection in SDN, listing the technique, the datasets used and the reported performance for a range of machine learning and hybrid models.

Table 1 Literature overview on ddos attack detection in software-defined networks

Author(s)	Technique/model	Key approach	Dataset(s)	Performance
Alashhab et al. [6]	Ensemble (DT, NB, perceptron)	Dynamic feature selection, real-time adaptation in SDN	Benchmark datasets, Mininet + Ryu	Avg. 99.2% detection rate
Hnamte et al. [7]	Deep neural network (DNN)	Scalable, real-time traffic monitoring	SDN, CICIDS2018, Kaggle DDoS	99.98%, 100%, 99.99% accuracy
Hussain et al. [8]	HRIDS (ML + rule-based)	Hybrid models on industrial traffic	F2F Industrial	99.97% accuracy, 99.96% TPR
Hassan et al. [9]	Entropy + K-means	Detection of DDoS, entropy-based traffic analysis combined with ML	CICIDS2017, CICIDS2019	Effective anomaly detection
Nisa et al. [10]	SVM, KNN (two-phase auth)	Packet filtration + ML	CICDoS2017, Mininet-Ryu	Reduced FPs, improved flow
Kavitha et al. [11]	ML in IoT-SDN	Flood attack mitigation via multi-controller	IoT-based cloud data	99.99% accuracy
Ouhssini et al. [12]	CNN-LSTM-transformer	Genetic algorithm + Auto CNN-DT	CIDDS-001	High real-time detection accuracy
Han et al. [13]	Feature selection + ML	Faster IDS for 5G SDN	CICIDS2017, CICIDS2018	Fast and accurate detection
Paidipati et al. [14]	MMEDRL-ADM	Metaheuristic + ensemble DRL	Benchmark datasets	High detection rate and efficiency
Chaudhary et al. [15]	RF + SU + K-means	Entropy + Chebyshev thresholding	BoT-IoT	98.71% accuracy, 35% lower latency
Santos-Neto et al. [16]	Hybrid ML-entropy	Dynamic entropy thresholding	DARPA, corporate	> 99% accuracy, considerable improvement
Kapourchali et al. [17]	P4httpGuard	ML + P4 switch offloading	Real-time P4 SDN	60 s faster, 81.89% less bandwidth
Abid et al. [18]	CNN + LSTM	Spatial–temporal traffic patterns	IoT SDN traffic	High accuracy, robust detection
John et al. [19]	PCA + Ensemble ML	Fuzzy C-Means + PCA + RF	WSN-DS	99.99% accuracy
Tonkal et al. [20]	NCA + DT, KNN, ANN, SVM	Feature selection + model testing	Public SDN DDoS dataset	100% accuracy (DT)

3 Methods

This section describes the datasets, classifiers and evaluation procedure used in the study. Section 3.1 presents the DDoS-SDN and APA-DDoS datasets and the flow level features they provide. Section 3.2 introduces the six supervised classifiers considered, namely K-Nearest Neighbours, Support Vector Machine, Random Forest, Decision Tree, Logistic Regression and Extreme Gradient Boosting, which are evaluated using accuracy, precision, recall, specificity, F1-score, ROC-AUC and PR-AUC within a leakage-controlled pipeline (Fig. 2). The aim of the comparison is to identify the classifiers that detect DDoS attacks in SDN most accurately and reliably.

Figure 2 shows the experimental pipeline. Identifier attributes are removed, and the records are split into stratified training (70%) and test (30%) portions with a fixed random seed of 42. Imputation, ordinal encoding, Mutual Information feature selection and standardisation are fitted on the training portion only and then applied unchanged to the test portion. Each classifier is trained once on the training portion and evaluated once on the held-out test set, and ten-fold cross-validation on the training portion repeats all of these steps inside every fold.

3.1 Dataset description and analysis

The first dataset is publicly available on Kaggle and was generated in a Mininet-based SDN environment for DDoS detection [21]. It contains 104,345 flow records described by 23 attributes that capture statistical flow metrics and network topology information, with benign and malicious traffic across the TCP, UDP and ICMP protocols. Each record is labelled as benign (0) or malicious (1). The timestamp (dt) and the source and destination IP addresses (src, dst) record when and between which hosts a flow was observed rather than how it behaves, and were therefore excluded from the model inputs in all experiments. The remaining 19 input

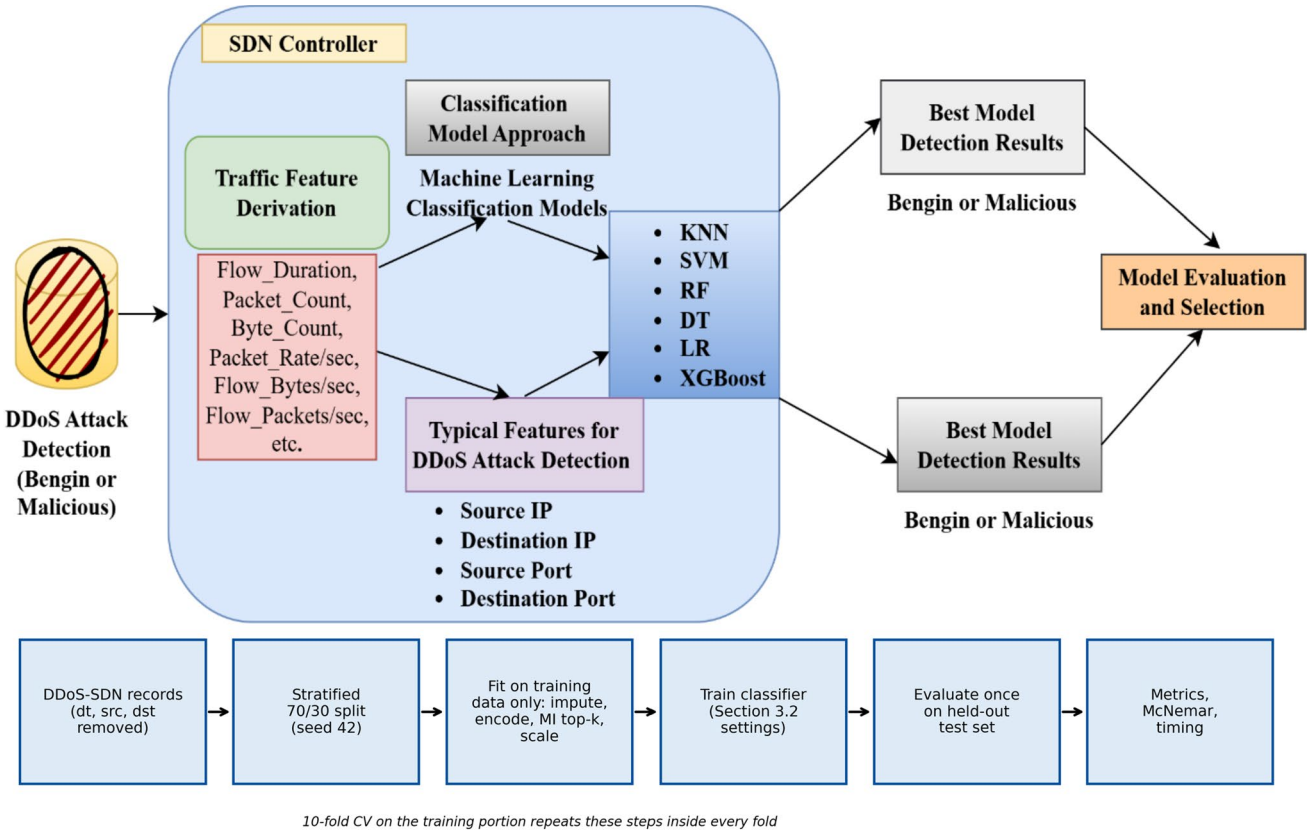


Fig. 2 Experimental workflow of DDoS attack detection

features are switch, pktcount, bytecount, dur, dur_nsec, tot_dur, flows, packetins, pktperflow, byteperflow, pktrate, Pairflow, Protocol, port_no, tx_bytes, rx_bytes, tx_kbps, rx_kbps and tot_kbps. The switch identifier and port number (switch, port_no) describe where a flow was observed in the emulated topology; they were retained in the main experiments, and their effect is tested by removing them in Sect. 4.8. After removal of 506 records with missing values, 103,839 records remained (63,335 benign and 40,504 attack), which were split by stratified sampling into 72,687 training and 31,152 test records.

The second dataset, APA-DDoS, was compiled to support the development of intrusion detection systems for contemporary networks. It complements existing public datasets by including attack patterns that they often omit, notably ACK flooding and PUSH-ACK flooding, which generate large volumes of malicious traffic in order to exhaust network resources and disrupt legitimate communication. The dataset contains both benign and malicious traffic described by packet-level header and traffic attributes, and it supports both binary and multiclass classification, and it was therefore initially considered as a second benchmark. The feature-level audit in Sect. 4.9 shows, however, that it is unsuitable for ranking classifiers, and it is reported as a dataset audit.

Table 2 summarises the APA-DDoS dataset. It comprises 151,200 packet-level records described by 23 attributes, of which 22 are candidate input attributes and the Label attribute serves as the class label; the attributes retained as model inputs are listed in Table 3. The dataset is designed for three-class classification and has a 2:1:1 ratio of one benign class to two attack classes and a large number of samples.

Table 3 lists the 22 input attributes of the APA-DDoS dataset and their treatment. Identifier fields that reflect how the traffic was generated in the testbed rather than attack behaviour, namely the source and destination IP addresses,

Table 2 Characteristics of the APA-DDoS dataset

Parameter	Value
Dataset name	APA-DDoS dataset
Total instances	151,200
Total features	23
Features used as inputs	22 (Raw-22); 9 after the audit (E1, Table 3)
target variable	Label
Number of classes	3
Dataset type	Multi-class classification

Table 3 APA-DDoS attributes and their treatment in the experiments

Attribute	Description	Treatment
ip.src	Source IP address	Excluded (identifier)
ip.dst	Destination IP address	Excluded (identifier; constant)
tcp.srcport	TCP source port	Excluded (identifier)
tcp.dstport	TCP destination port	Excluded (identifier; constant)
ip.proto	IP protocol number	Excluded (constant)
frame.len	Frame length (bytes)	Used
tcp.flags.syn	TCP SYN flag	Excluded (constant)
tcp.flags.reset	TCP RST flag	Excluded (constant)
tcp.flags.push	TCP PSH flag	Used (removed in ablation E2)
tcp.flags.ack	TCP ACK flag	Excluded (constant)
ip.flags.mf	IP More-Fragments flag	Excluded (constant)
ip.flags.df	IP Don't-Fragment flag	Used
ip.flags.rb	IP reserved bit	Excluded (constant)
tcp.seq	TCP sequence number	Excluded (identifier; constant)
tcp.ack	TCP acknowledgement number	Excluded (identifier; constant)
frame.time	Packet capture timestamp	Excluded (identifier)
packets	Packets in flow	Used
bytes	Bytes in flow	Used
tx packets	Transmitted packets	Used
tx bytes	Transmitted bytes	Used
rx packets	Received packets	Used
rx bytes	Received bytes	Used

Table 4 Class distribution

Class label	Number of instances
Benign traffic	75,600
DDoS-PSH-ACK	37,800
DDoS-ACK	37,800
total	151,200

Table 5 Dataset split used in experiments

Dataset portion	Percentage	Number of instances
Training set	70%	105,840
Testing set	30%	45,360
Total	100%	151,200

source and destination ports, the packet timestamp (frame.time) and the TCP sequence and acknowledgement numbers, were excluded from the model inputs. Attributes that are constant across the training set were also removed, leaving nine input features. In the Raw-22 configuration all 22 attributes are used as inputs; the effect of including the identifier fields is analysed in Sect. 4.9.

Table 4 reports the class distribution of the APA-DDoS dataset. Of the 151,200 samples, 75,600 are benign, 37,800 are DDoS-PSH-ACK and 37,800 are DDoS-ACK. This distribution provides sufficient benign and attack traffic to evaluate multiclass DDoS detection.

Table 5 summarises the partitioning of the APA-DDoS dataset. The 151,200 samples were split by stratified sampling (seed 42) into a training set of 70% (105,840 samples), used to learn the model parameters, and a test set of 30% (45,360 samples), used to assess performance on unseen data. This split supports a reliable assessment of generalisation.

3.2 Machine learning in DDoS attacks

A single preprocessing and evaluation pipeline was used for both datasets as shown in Fig. 2. Records with missing values were removed, and the data were split by stratified sampling into training (70%) and test (30%) portions with a fixed random seed of 42. Median imputation, ordinal encoding of categorical attributes (with categories unseen in training mapped to -1), Mutual Information feature selection and standardisation to zero mean and unit variance were implemented as one scikit-learn pipeline, fitted on the training portion only and then applied unchanged to the test portion; feature scaling is particularly important for the distance-based classifiers KNN and SVM. Mutual

Information was estimated with the k-nearest-neighbour estimator (three neighbours, random state 42) on the training portion, and the Top-5 and Top-12 subsets were formed from the resulting ranking. Five features represent an aggressive reduction aimed at latency-critical deployment, and twelve features a moderate reduction that removes the seven lowest-ranked features. Both sizes were fixed in advance rather than tuned on the test set. The classifiers used the following fixed hyperparameters:

- Logistic Regression with the liblinear solver (L-BFGS with a multinomial formulation, Eq. (11), for the three-class APA-DDoS task), a regularisation parameter C of 0.03 and at most 1,000 iterations;
- SVM with a radial basis function kernel and C = 1.0; KNN with five neighbours, Manhattan distance and distance-based weighting;
- Decision Tree with the Gini criterion, a maximum depth of 8 and at most 11 leaf nodes;
- Random Forest with 20 trees, entropy splitting, a maximum depth of 10, at least 100 samples per split and 20 per leaf, and square-root feature sampling, and;
- XGBoost with log-loss as the evaluation metric and otherwise default hyperparameters.

These constrained settings favour small and fast models. To test whether they limit the tree-based models, an unconstrained Decision Tree, a Random Forest of 200 fully grown trees and an XGBoost model with 500 trees of depth 10 were also evaluated (Sect. 5.3). All hyperparameters were fixed a priori and were not tuned on the test set, and the test set was used once, for the final evaluation. Stratified ten-fold cross-validation was carried out on the training portion only, with the whole pipeline refitted in every fold. SVM was excluded from cross-validation because ten RBF-kernel fits per feature set on about 65,000 records each were computationally impractical, and it is evaluated on the held-out test set only. All experiments were run on a CPU only (Intel Xeon 2.20 GHz, four logical cores, 31.3 GB RAM; Python 3.13, scikit-learn 1.6.1 and XGBoost 3.4.1).

The DDoS detection pipeline used here combines traffic analysis with supervised classification to support timely detection in SDN environments. Machine learning is well-suited to DDoS detection because it can learn discriminative patterns from large volumes of network traffic without requiring hand-crafted signatures. In SDN environments, packet counts, byte counts, and flow duration are examples of flow-based features used as statistical signals that supervised classifiers exploit to separate benign and malicious traffic. The classifiers in this study are trained on historical traffic data and tested for their ability to detect DDoS attacks. Each of the classifiers is introduced individually in the rest of this section.

3.2.1 K-nearest neighbors

KNN is a non-parametric, instance-based classifier, which makes it well suited to anomaly detection in SDN. The SDN controller collects real-time flow statistics from OpenFlow-enabled switches, such as packet count, packet rate and flow duration, and represents each flow as a feature vector. A test flow is classified by computing its distance to the labelled training instances and assigning the majority class among its K nearest neighbours. Although KNN can be computationally expensive on large datasets, it is simple and often accurate, particularly when combined with feature or dimensionality reduction. The class membership probability is given by:

$$P(y = jX = x) = \frac{1}{K} \sum_{i \in A} I\left(y^{(i)} = j\right) \tag{1}$$

where K is the number of nearest neighbors, A is the index set of the K nearest neighbors of x,$y^{(i)}$ is the class label of the i-th neighbor, and I(·) is the indicator function that returns 1 if the condition is true, else 0.

3.2.2 Support vector machine

SVM is a discriminative classifier that finds the maximum margin hyperplane separating legitimate from malicious traffic in a high-dimensional feature space. In SDN, the controller aggregates flow level information from OpenFlow-enabled switches, including flow size in bytes or packets, flow duration and flow rate, and passes the resulting feature vectors to the model. For data that are not linearly separable, kernel functions such as the radial basis function and polynomial kernels map the features into a higher dimensional space in which the classes become more easily separable, which improves detection. The SVM decision function is:

$$f(x) = \text{sign}\left(\sum_{i=1}^{n} \alpha_i y_i K\left(x_i, x\right) + b \right) \tag{2}$$

where x is the input feature vector, x_i are support vectors, $y_i \in \{-1, +1\}$ are class labels, α_i are Lagrange multipliers, $K(\cdot, \cdot)$ is the kernel function, e.g., linear and RBF, and b is the bias term. The centralized SDN architecture enables real-time feature aggregation and dynamic mitigation via flow rules based on SVM-derived classification results.

3.2.3 Logistic regression

Logistic Regression is a linear, probabilistic model widely used for binary classification tasks such as DDoS detection in SDN. Using flow-based features collected by the

controller, such as packet count, flow duration and byte rate, it estimates the conditional probability of the class label y ∈ {0, 1} for an input feature vector x through the sigmoid function:

$$P(y = 1X) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}} \quad (3)$$

where $x = (x_1, x_2, \ldots, x_n)$ is the input feature vector, β_0 is the intercept, and $\beta_1, \beta_2, \ldots, \beta_n$ are the regression coefficients estimated from the training data by maximum likelihood. A flow is classified as an attack ($y = 1$) when $P(y = 1 \mid x)$ exceeds a decision threshold, typically 0.5, and as benign ($y = 0$) otherwise.

3.2.4 Decision tree

DT is a non-parametric supervised learning method used for classification and regression. It works by recursively splitting the dataset based on features that provide the highest information gain for classification or minimum variance for regression. In classification tasks, a DT selects the best feature at each node using a splitting criterion, such as the Gini Index, Information Gain, or Classification Error. In SDN-based DDoS detection, the controller collects flow-based features such as packet count, flow duration, and byte rate, and the tree learns to partition the feature space into regions of benign and malicious traffic. The tree is built recursively by selecting the feature that maximizes class separation, producing a hierarchical decision model. At each decision node, the classifier selects the feature that maximizes the Information Gain (IG), defined as Eq. (4)

$$IG(D, A) = H(D) - \sum_{v \in \text{Values}(A)} \frac{|D_v|}{|D|} H(D_v) \quad (4)$$

where H(D) is the entropy of the dataset D, Values(A) is the set of distinct values A can take, $D_v \subseteq D$ is the subset of records for which A = v, and $|\cdot|$ denotes set cardinality. The entropy H(D) is calculated by:

$$H(D) = -\sum_{i=1}^{C} p_i \log_2(p_i) \quad (5)$$

where C is the number of classes and p_i is the proportion of samples in D belonging to class i.

3.2.5 Random forest

Random Forest is an ensemble method that combines many Decision Trees to improve accuracy and robustness. Each tree is trained on a bootstrap sample of the data, and at each node a random subset of features is considered for splitting. This combination of bagging and feature randomness reduces variance and limits overfitting. For DDoS detection in SDN, the controller collects flow statistics such as packet count, flow duration and byte rate, and the forest assigns each flow to the class chosen by a majority vote of its trees. The classification rule is:

$$\hat{y}(x) = \underset{c \in C}{\arg\max} \sum_{t=1}^{T} I(h_t(x) = c) \quad (6)$$

where x is the input feature vector, $\hat{y}(x)$ is the predicted class label, T is the number of decision trees in the forest, $h_t(x)$ is the class predicted by the t-th tree, C is the set of class labels, and $I(\cdot)$ is the indicator function that returns 1 if its argument is true and 0 otherwise. The forest output is the class receiving the most votes across the T trees.

3.2.6 Extreme gradient boosting (XGBoost)

XGBoost is an ensemble method that builds an additive model of Decision Trees in a stagewise manner. For DDoS detection in SDN, it processes flow-based features such as flow duration, packet count and bytes per second and outputs a score indicating whether a flow is benign or malicious. Its training objective combines a loss term that measures the fit to the training data with a regularisation term that penalises model complexity:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \quad (7)$$

where $l(y_i, \hat{y}_i)$ is the per-sample loss between the true label y_i and the prediction $\hat{y}_i$, n is the number of training samples, K is the number of trees in the ensemble, and $\Omega(f_k)$ is the regularization term applied to the k-th tree f_k. The regularisation term is defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \quad (8)$$

where T is the number of leaves in a tree, w is the vector of leaf weights, and γ and λ control the penalties on tree size and on leaf weight magnitude, respectively. Combining gradient boosting with explicit regularisation gives XGBoost strong generalisation on high dimensional traffic data while limiting overfitting, which motivates its inclusion in this comparison.

3.2.7 Binary and multiclass classification setups

For the multiclass setting, the decision functions of SVM and Logistic Regression are given in Eqs. (9-12). In the case of SVM, the model computes a score or probability for every class and assigns the input to the class with the highest value:

$$f_k(x) = \sum_{i \in SV_k} \alpha_{ik} y_i K(x_i, x) + b_k, k = 1,2,\ldots,K \quad (9)$$

The predicted class is:

$$\hat{y} = \arg \max_{k \in \{1,\ldots,K\}} f_k(x) \quad (10)$$

For multiclass Logistic Regression we have:

$$P(y = k \mid x) = \frac{\exp(w_k^T x + b_k)}{\sum_{j=1}^{K} \exp(w_j^T x + b_j)}, k = 1,2,\ldots,K \quad (11)$$

And the predicted class can also be written as:

$$\hat{y} = \arg \max_{k \in \{1,\ldots,K\}} P(y = k \mid x) \quad (12)$$

The two datasets correspond to different classification settings. The DDoS-SDN dataset defines a binary problem (benign versus attack), whereas the APA-DDoS dataset defines a three-class problem (Benign, DDoS-ACK and DDoS-PSH-ACK). The classifiers are adapted to each setting as follows. Logistic Regression uses the sigmoid function with binary cross-entropy loss in the binary case and a softmax (multinomial) formulation with categorical cross-entropy loss in the multiclass case, as given in Eqs. (11) and (12). The support vector machine uses a hinge loss margin in the binary case and a one-vs-one scheme for the three-class case, in which a separate classifier is trained for each pair of classes and the final label is decided by majority vote, as given in Eqs. (9) and (10). XGBoost minimises the regularised logistic objective for binary classification and the softmax objective with multiclass log-loss for the three-class case. The Decision Tree, Random Forest and KNN are inherently multiclass and need no such adaptation. For the multiclass results on APA-DDoS, precision, recall and F1-score are reported with macro, micro and weighted averaging, as reported for APA-DDoS in Sect. 4.9. Macro averaging treats all classes equally, micro averaging aggregates over all samples, and weighted averaging accounts for the number of samples in each class.

4 Results and discussion

This section defines the evaluation metrics and presents the results. Section 4.2 audits both datasets, Sect. 4.3 reports the Mutual Information ranking, Sects. 4.4–4.6 report the DDoS-SDN test-set results with all 19 features and with the MI Top-5 and Top-12 subsets, Sect. 4.7 reports ten-fold cross-validation, Sect. 4.8 tests robustness under stricter splits, and Sect. 4.9 reports the APA-DDoS results and feature audit. All scores are computed on held-out test data unless stated otherwise.

4.1 DDoS attack performance metrics

Seven metrics are used to evaluate the models, namely accuracy, precision, recall, specificity, F1-score, the area under the ROC curve (ROC-AUC) and the area under the precision–recall curve (PR-AUC). Together they describe each model's ability to distinguish benign from malicious traffic, including the balance between missed attacks and false alarms.

Table 6 defines the confusion-matrix-based metrics and the two threshold-free measures. For the binary DDoS-SDN task, precision, recall, specificity and F1-score refer to the attack class; for the three-class APA-DDoS task they are macro-averaged over the classes, and ROC-AUC and PR-AUC are computed one-versus-rest and macro-averaged.

4.2 Dataset audit and descriptive statistics

Table 7 summarises the descriptive statistics of the 19 DDoS-SDN input features computed on the 103,839 records used. The features span very different ranges; for example, bytecount reaches 1.47×10^8 and tot_dur 1.88×10^{12}, whereas Pairflow is binary, which is why all features are standardised before classification. Figure 3 shows the correlation matrix of the input features: dur and

Table 6 Evaluation metrics definitions and formulas

Metric	Formula	Definition
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Measures the overall correctness of the model's predictions
Precision	TP / (TP + FP)	Indicates the proportion of correctly predicted abnormal traffic among all traffic predicted as abnormal
Recall	TP / (TP + FN)	Measures the model's effectiveness in detecting malicious network traffic
F1-Score	(2 × TP) / (2 × TP + FP + FN)	Harmonic mean of Precision and Recall, providing a balanced measure of model performance
Specificity	TN / (TN + FP)	Proportion of benign traffic correctly recognised as benign (true negative rate)
ROC-AUC	Area under the TPR–FPR curve	Threshold-free measure of how well the model ranks attack traffic above benign traffic
PR-AUC	Area under the precision–recall curve	Threshold-free measure that emphasises performance on the attack class

tot_dur are perfectly correlated (r = 1.00), and pktperflow and byteperflow (r = 0.82) and pktcount and bytecount (r = 0.68) are strongly correlated, so several features carry overlapping information.

Table 8 audits both datasets before any model is trained. The DDoS-SDN dataset contains 5,091 exact duplicate records (4.9%) but no feature vector with conflicting labels, so a perfect classifier is possible in principle, and 6.55% of the test records have an identical feature vector in the training portion. The APA-DDoS dataset has no duplicate records when all 22 attributes are used, because frame time is unique to every packet. Once the identifier and constant attributes are removed (E1, nine features), the 151,200 records reduce to 65 distinct feature vectors, all of which occur in both the training and the test portions. Removing the TCP PSH flag as well (E2, eight features) leaves 44 distinct vectors, 21 of which occur with more than one label, which limits the attainable accuracy to 75.78%.

4.3 Mutual-information feature ranking

Figure 4 shows the MI score of every DDoS-SDN input feature estimated on the training portion, and Figs. 5, 6, and 7 present the MI heatmap and the most and least informative features. The volume features dominate the ranking: bytecount (0.603), pktcount (0.582), byteperflow (0.556), pktperflow (0.541) and pktrate (0.385) form the Top-5 subset. The Top-12 subset adds tot_dur, dur, packetins, tx_bytes, rx_bytes, tot_kbps and rx_kbps. The topology attributes switch (0.005) and port_no (0.003) carry almost no information about the label, as do Pairflow (0.010) and flows (0.029). The ranking is computed inside the training pipeline, so the test set does not influence which features are selected.

4.4 Classification with All 19 features

Table 9 reports the test-set performance of the six classifiers with all 19 input features, and Fig. 8 shows the corresponding confusion matrices. XGBoost classifies all 31,152 test records correctly, and Random Forest reaches 99.47% accuracy with 122 false positives and 43 false negatives. KNN (98.47%) and the Decision Tree (98.20%) follow, and SVM reaches 96.86%. Logistic Regression is clearly the weakest model at 76.64% accuracy and misses 4,098 of the 12,151 attacks (recall 66.27%), which indicates that the two classes are not linearly separable in the standardised feature space. Figure 9 shows the ROC curves: every non-linear classifier attains a ROC-AUC of at least 0.992, whereas Logistic Regression reaches 0.838.

4.5 Classification with the MI Top-5 features

Table 10 reports the test-set results with the five highest-ranked features, together with the change in accuracy relative to the full feature set, and Fig. 10 shows the confusion matrices. Reducing the input from 19 to 5 features (a 73.7% reduction) lowers the accuracy of most classifiers: by 5.80 percentage points for Logistic Regression, 4.48 for the Decision Tree, 4.32 for SVM, 2.57 for Random Forest and 0.95 for XGBoost. KNN is the exception: its accuracy rises from 98.47% to 99.97%, with 9 false positives and no missed

Table 7 Descriptive statistics of the DDoS-SDN input features

Feature	Mean	SD	25%	Median	75%	Max
Switch	4.21	1.96	3	4	5	10
pktcount	52,781.70	52,061.41	801	42,557	94,683	260,006
bytecount	3.80×10^7	4.87×10^7	79,282	6.41×10^6	7.60×10^7	1.47×10^8
dur	322.13	283.93	128	252	413	1,881
dur_nsec	4.61×10^8	2.77×10^8	2.33×10^8	4.18×10^8	7.04×10^8	9.99×10^8
tot_dur	3.23×10^{11}	2.84×10^{11}	1.28×10^{11}	2.52×10^{11}	4.13×10^{11}	1.88×10^{12}
flows	5.67	2.95	3	5	7	17
packetins	5,196.26	5,266.63	1,943	3,024	7,385	25,224
pktperflow	6,365.87	7,413.26	29	8,304	10,004	19,190
byteperflow	4.69×10^6	7.56×10^6	2,842	544,736	9.73×10^6	1.50×10^7
pktrate	211.68	247.14	0	276	333	639
Pairflow	0.60	0.49	0	1	1	1
port_no	2.34	1.08	1	2	3	5
tx_bytes	9.37×10^7	1.52×10^8	4,799	4.55×10^6	1.36×10^8	1.27×10^9
rx_bytes	9.37×10^7	1.33×10^8	3,539	1.40×10^7	1.44×10^8	9.91×10^8
tx_kbps	1,003.77	2,428.36	0	0	255	20,580
rx_kbps	1,003.81	2,054.89	0	0	557	16,577
tot_kbps	2,007.58	3,144.44	0	4	3,838	20,580

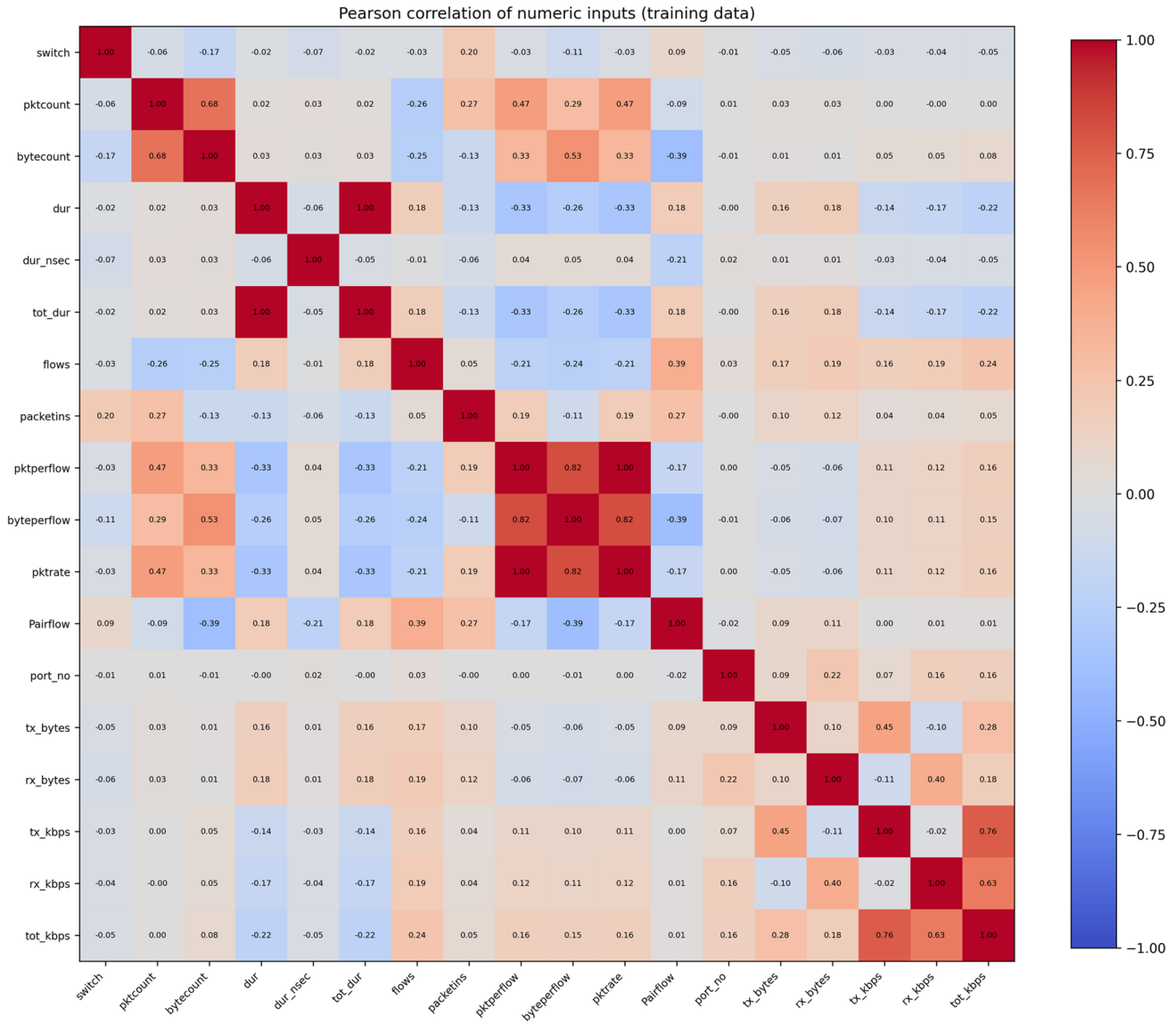


Fig. 3 Correlation matrix of the DDoS-SDN input features

Table 8 Dataset audit of DDoS-SDN and of APA-DDoS under three feature sets

Quantity	DDoS-SDN (19 features)	APA Raw-22	APA E1 (9 features)	APA E2 (8 features)
Records used	103,839	151,200	151,200	151,200
Records removed (missing values)	506	0	0	0
Duplicate records (features + label)	5,091	0	151,135	151,135
Distinct feature vectors	98,748	151,200	65	44
Vectors with more than one label	0	0	0	21
Accuracy ceiling (%)	100	100	100	75.78
Test records whose vector occurs in training (%)	6.55	0	100	100

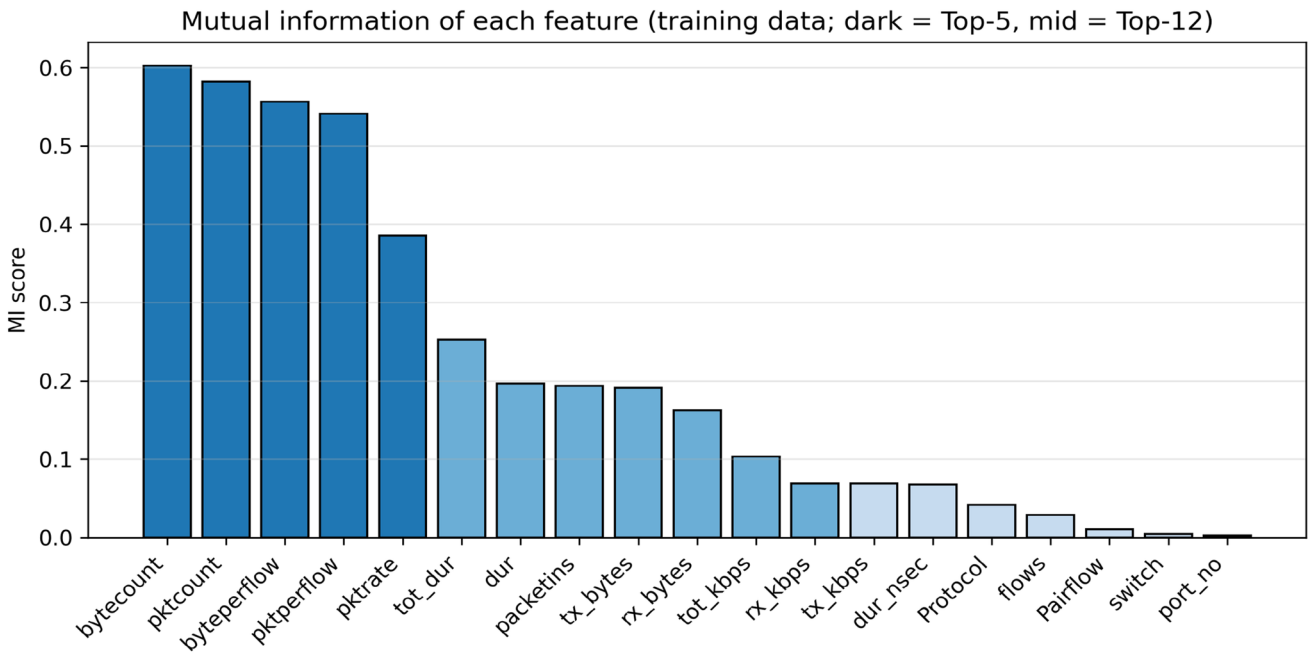


Fig. 4 Mutual information score of each DDoS-SDN input feature (training portion)

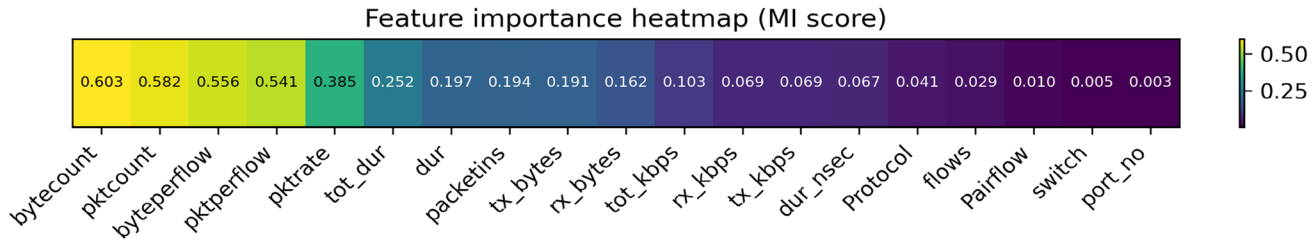


Fig. 5 Heatmap of mutual information scores of the DDoS-SDN input features

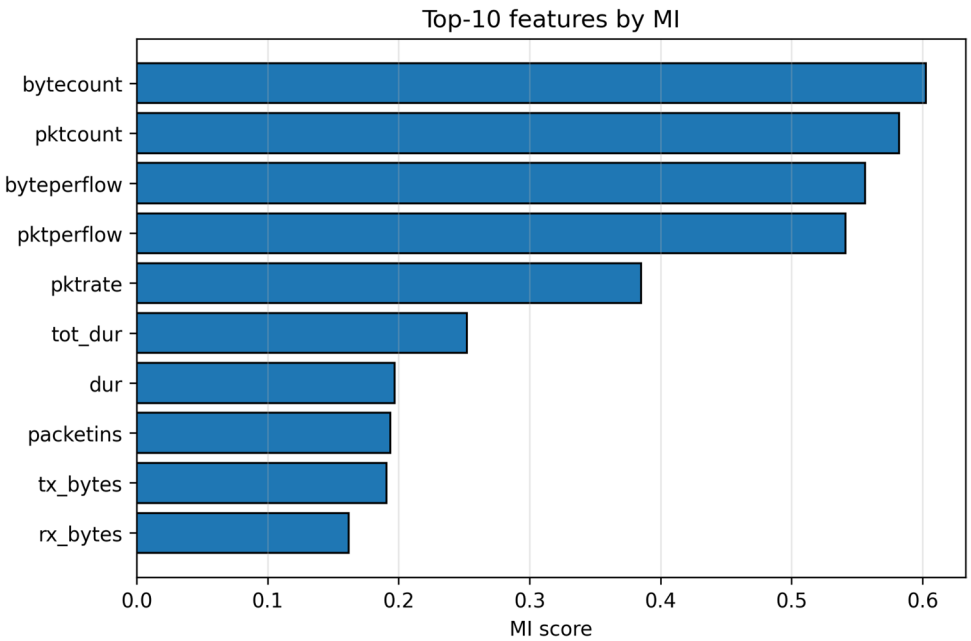


Fig. 6 Ten most informative DDoS-SDN features by mutual information

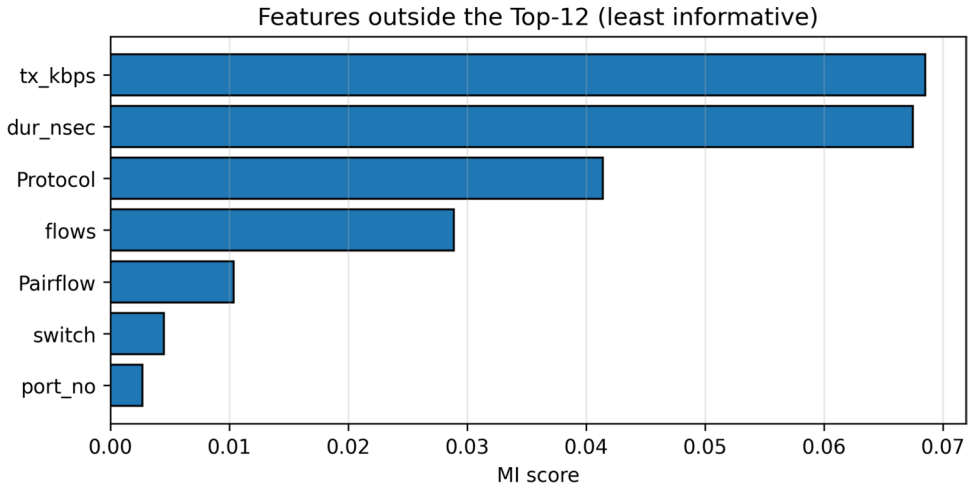


Fig. 7 Least informative DDoS-SDN features by mutual information

Table 9 Test-set performance on DDoS-SDN with all 19 features (%, attack class; ROC-AUC and PR-AUC as fractions)

Model	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	PR-AUC
Logistic regression	76.64	71.70	66.27	83.27	68.88	0.8375	0.7552
SVM	96.86	95.25	96.77	96.92	96.00	0.9947	0.9906
KNN	98.47	98.15	97.91	98.82	98.03	0.9974	0.9951
Decision tree	98.20	96.70	98.76	97.85	97.72	0.9924	0.9773
random forest	99.47	99.00	99.65	99.36	99.32	0.9999	0.9998
XGBoost	100.00	100.00	100.00	100.00	100.00	1.0000	1.0000

attacks, because distances computed on the five-volume features are no longer diluted by weakly informative dimensions. Figures 11, 12, 13, and 14 compare the full and Top-5 feature sets in terms of accuracy and training time, change in weighted F1-score, total computational cost and a combined accuracy and F1-score heatmap. The largest efficiency gain is again for KNN, whose inference time on the test set falls from 23.66 s to 0.21 s.

4.6 Classification with the MI Top-12 features

Table 11 and Fig. 15 report the results with the twelve highest-ranked features (a 36.8% reduction). The Top-12 subset preserves nearly the full accuracy of the tree-based classifiers: XGBoost reaches 99.99% (two missed attacks), Random Forest 99.40% and the Decision Tree 98.23%, all within 0.07 percentage points of the full feature set, and KNN improves from 98.47% to 99.31%. SVM and Logistic Regression lose 0.97 and 2.25 percentage points, respectively. Figures 16, 17, 18, and 19 present the corresponding accuracy, F1-score, cost and heatmap comparisons. Top-12 is therefore the better choice when detection accuracy must be preserved, whereas Top-5 gives the largest cost reduction at a moderate loss of accuracy for most classifiers.

4.7 Ten-fold cross-validation

Tables 12, 13, and 14 report stratified ten-fold cross-validation on the training portion for the five classifiers other than SVM, with the whole pipeline, including MI selection, refitted in every fold, and Fig. 20 shows the mean accuracies with their standard deviations. The fold-to-fold standard deviation of accuracy is at most 0.61 percentage points, and the cross-validated means differ from the corresponding held-out test accuracies by at most 0.47 percentage points, so the test-set results are not an artefact of a single favourable split. For every feature set, the classifiers are ranked in the same order in cross-validation as on the test set.

4.8 Robustness under stricter splits

The random split places flows recorded at the same time and between the same hosts in both the training and the test portions. Table 15 and Fig. 21 therefore report three stricter evaluations with the same pipeline and hyperparameters. In the time-ordered split, the earliest 70% of the records of each class (ordered by the dt timestamp) are used for training and the latest 30% for testing. In the unseen-host split, all flows from four of the 19 source hosts (10.0.0.1, 10.0.0.5, 10.0.0.9 and 10.0.0.12; 30.0% of the records) are held out for testing. In the third evaluation, the topology attributes switch and

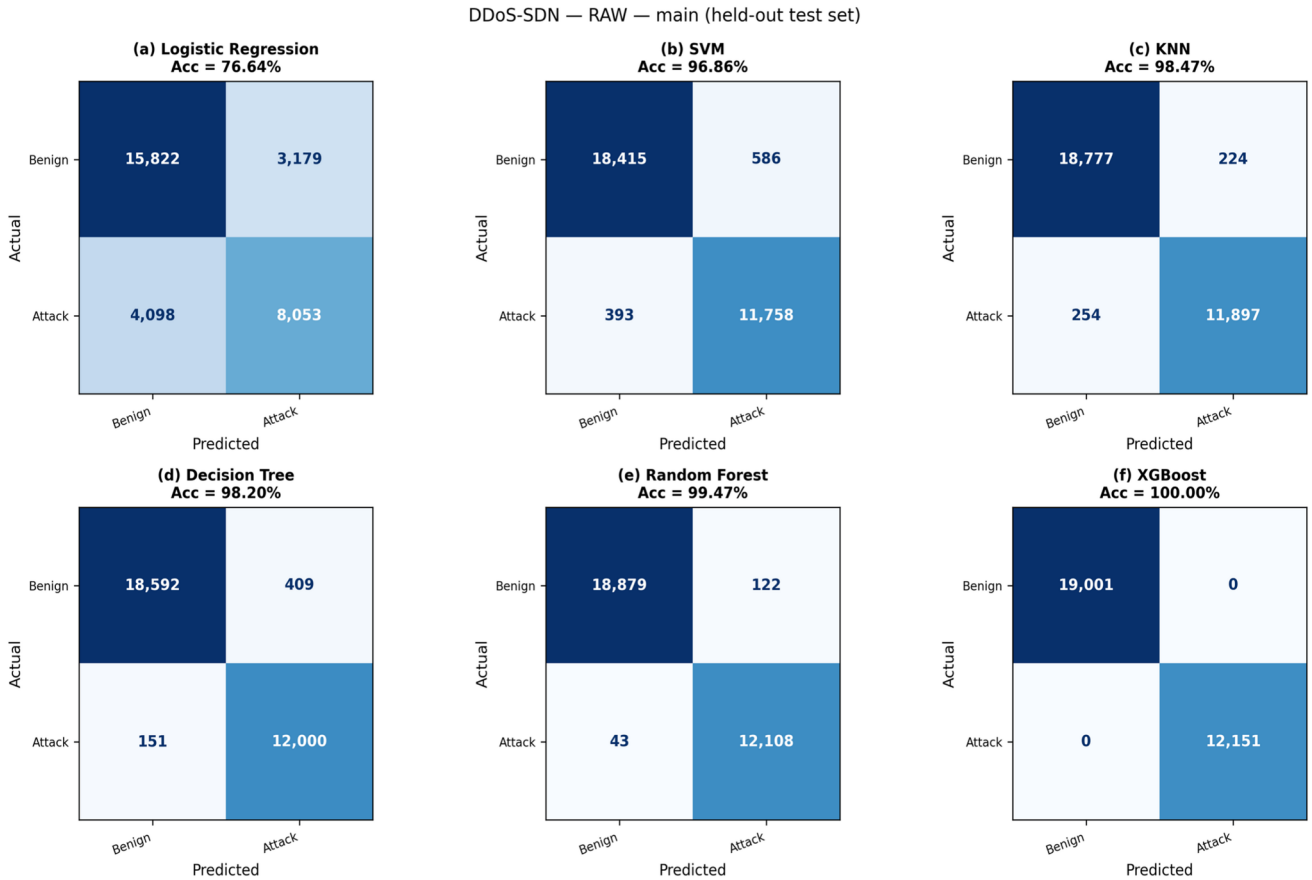


Fig. 8 Confusion matrices of the six classifiers on the DDoS-SDN test set with all 19 features

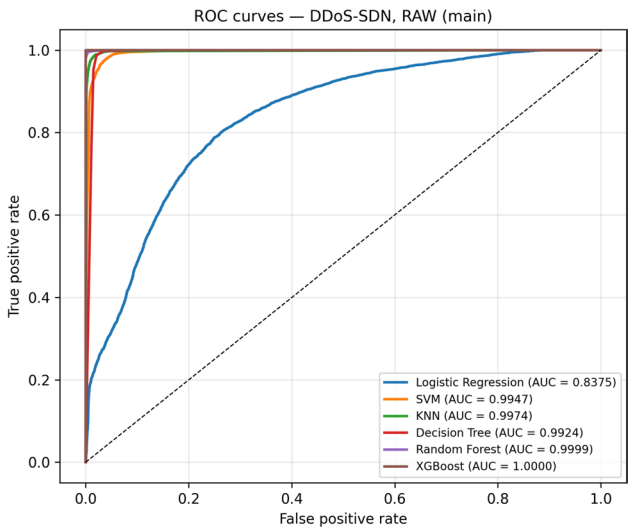


Fig. 9 ROC curves of the six classifiers on the DDoS-SDN test set with all raw features

port_no are removed and the random split is kept. Removing the topology attributes changes accuracy by at most 0.57 percentage points, so the results do not depend on where a flow was observed. Holding out source hosts lowers accuracy by up to 4.86 percentage points (KNN), while Random Forest and XGBoost remain above 98%. The time-ordered split has the largest effect: with all 19 features, accuracy falls by between 6.67 (XGBoost) and 17.76 (Decision Tree) percentage points, attack recall falls to 75–90% for the non-linear classifiers, and with the Top-12 subset the drop for XGBoost reaches 20.13 percentage points. Random-split accuracies close to 100% should therefore be read as an upper bound for traffic recorded under the same conditions, not as an estimate of performance on later traffic.

4.9 Results and feature audit on the APA-DDoS dataset

The APA-DDoS dataset was initially considered as a second benchmark for ranking the classifiers, but its results proved to depend on how a small number of attributes were handled rather than on detection ability. This section therefore reports the APA-DDoS results and then audits the dataset at the feature level to explain them. Section 4.9.1 presents the test set performance with the original 22 attributes and the MI subsets, and Sect. 4.9.2 traces the behaviour to identifier

Table 10 Test-set performance on DDoS-SDN with the MI Top-5 features

Model	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	ΔAcc
Logistic regression	70.84	65.94	52.21	82.75	58.28	0.7655	−5.80
SVM	92.54	90.26	90.66	93.74	90.46	0.9739	−4.32
KNN	99.97	99.93	100.00	99.95	99.96	1.0000	+1.51
Decision tree	93.72	86.84	98.90	90.42	92.48	0.9807	−4.48
Random forest	96.90	94.55	97.68	96.40	96.09	0.9967	−2.57
XGBoost	99.05	98.34	99.24	98.93	98.79	0.9996	−0.95

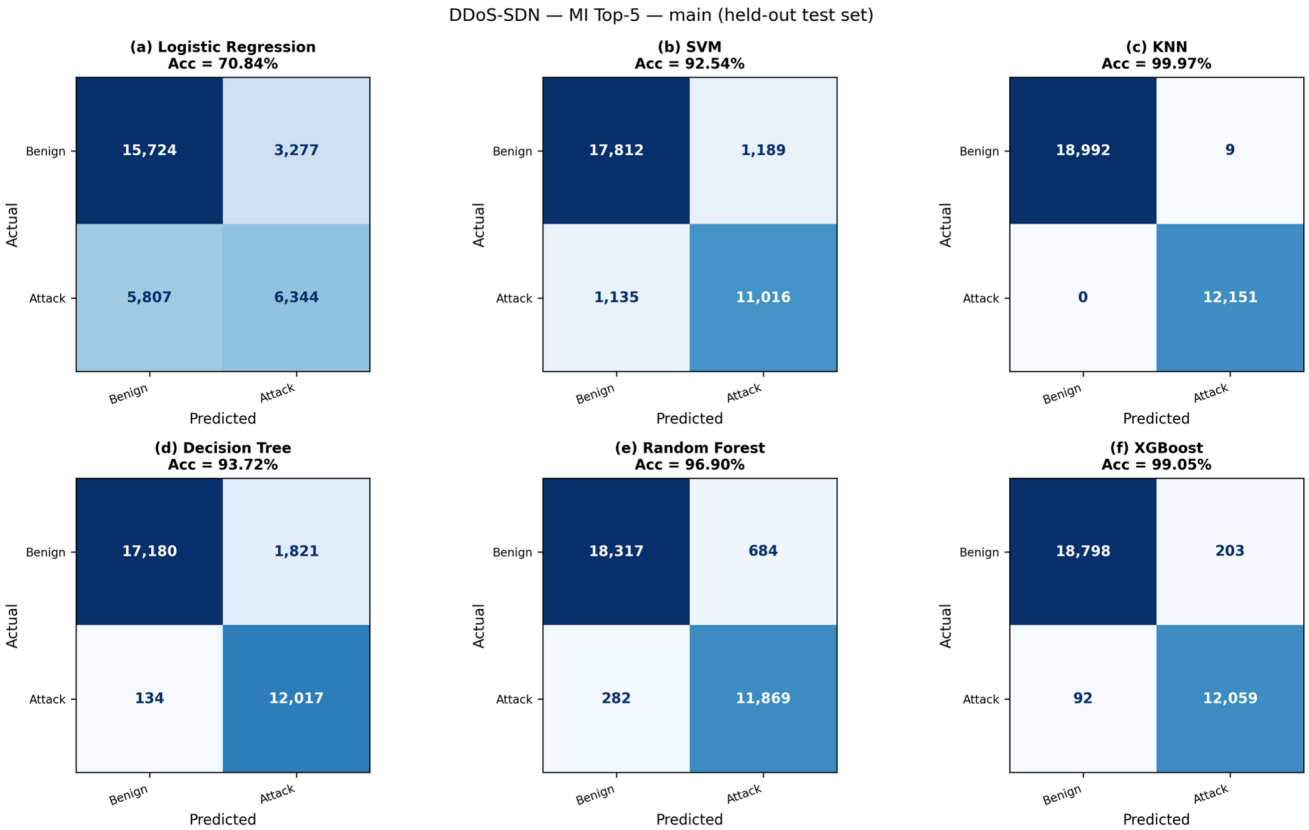


Fig. 10 Confusion matrices of the six classifiers on the DDoS-SDN test set with the MI Top-5 features

leakage from the capture timestamp, the definition of the two attack classes by the TCP PSH flag, and the small number of distinct feature vectors that remain once the identifier and constant attributes are removed.

4.9.1 Results with the original attributes and MI subsets

Table 16 reports the APA-DDoS test-set results for the original 22 attributes (Raw-22) and for the MI Top-5 and Top-12 subsets selected from them, and Fig. 23 shows the Raw-22 confusion matrices. Figure 22 shows the MI ranking: the capture timestamp frame.time has by far the highest score (1.04), and the identifiers ip.src and tcp.srcport also enter the Top-12 subset. With Raw-22, Logistic Regression, SVM and KNN reach 99.73–100% accuracy, whereas the Decision Tree, Random Forest and XGBoost reach only 75.00–77.57%, because they assign almost every DDoS-ACK packet to DDoS-PSH-ACK (DDoS-ACK recall of 0%, 6.27% and 10.30%, respectively). The Top-5 subset (frame.time, ip.flags.df, tx bytes, frame.len and bytes) contains no attribute that separates the two attack classes, and five of the six classifiers produce identical predictions at 75.00% accuracy. The Top-12 subset includes tcp.flags.push, and the Raw-22 pattern returns, with Random Forest at 95.07%. These inconsistent rankings, which depend on how each model handles the timestamp, motivated the feature-level audit below. The Raw-22 and Top-12 figures in Table 16 therefore reflect this leakage and are not valid classifier

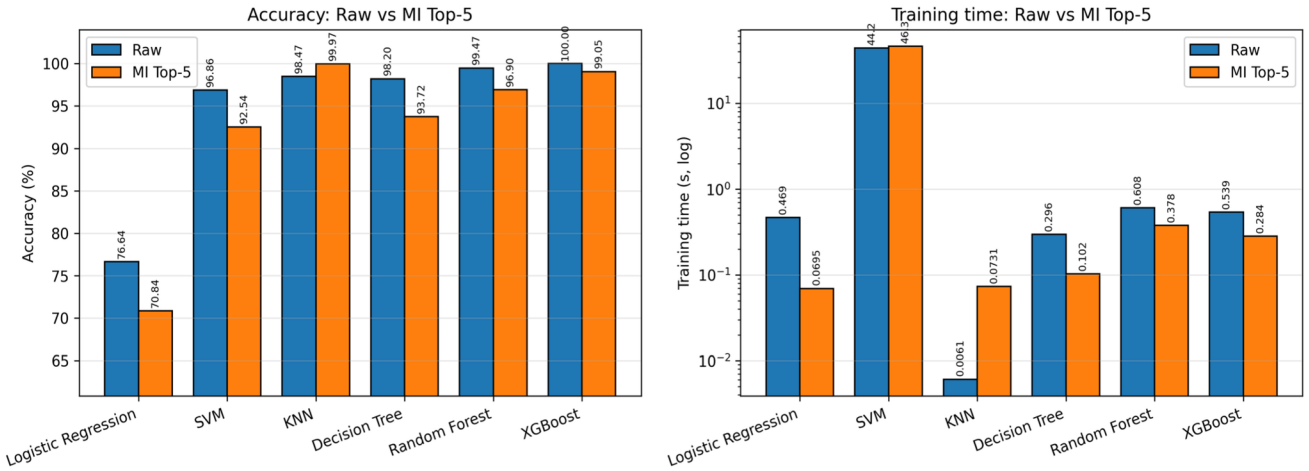


Fig. 11 Test accuracy and training time with all features and with top-5 features

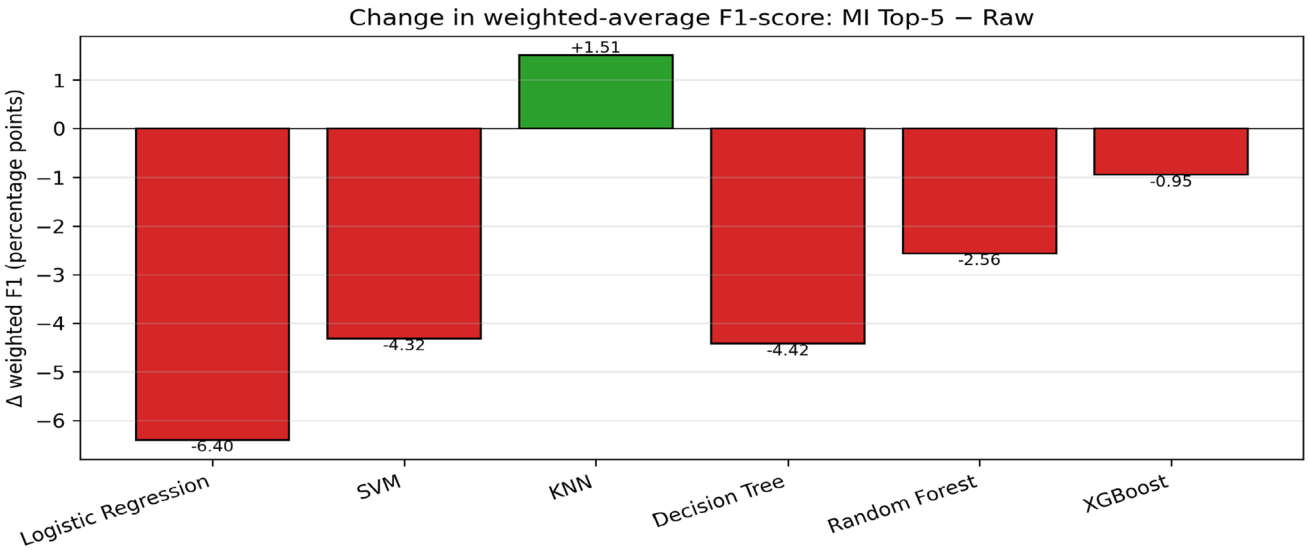


Fig. 12 Change in weighted F1-score after MI Top-5 feature selection

results. The audited outcome, with the identifier fields removed, is reported in Sect. 4.9.2 (Table 17) (Fig. 23).

4.9.2 Feature-level audit

Table 3 lists the 22 APA-DDoS attributes and their treatment. Two mechanisms explain the Raw-22 results. First, frame.time is unique to every packet, so after ordinal encoding on the training portion every test timestamp is an unseen category and is mapped to the same value (−1). The tree-based models, which split on frame.time, route all of these records down the same branches and therefore predict a single attack class for both attack types, whereas the linear and distance-based models rely on the remaining attributes, including the PSH flag. In an earlier run of the same pipeline, with the same split and settings, in which frame.time was converted into a continuous time value (E0b), every classifier reached 100% test accuracy, because the attacks were captured at different times and the timestamp therefore encodes the class label. Second, the two attack classes are defined by the TCP PSH flag: a one-split decision stump on tcp.flags.push separates DDoS-ACK from DDoS-PSH-ACK with 100% accuracy, whereas the next-best single attribute, tcp.srcport, reaches 51.89%, close to chance.

Table 17 and Figs. 24, 25, and 26 report the audit configurations. E1 removes the identifier and constant attributes and keeps nine features; every classifier, constrained or unconstrained, reaches 100% accuracy (Fig. 24). This reflects the redundancy of the data rather than detection

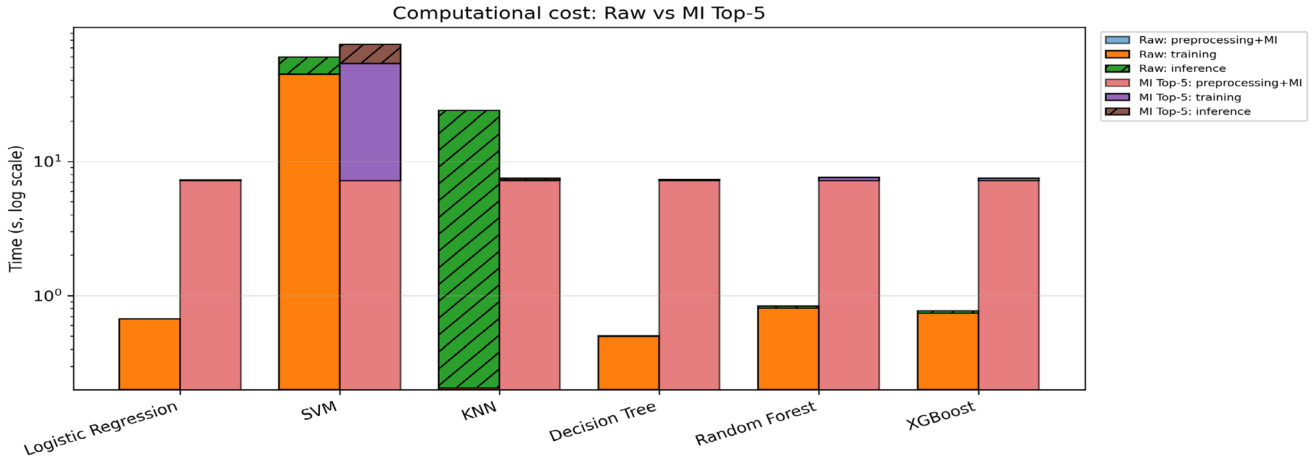


Fig. 13 Computational cost (preprocessing and MI fitting, training and inference) with all features and with the MI Top-5 features

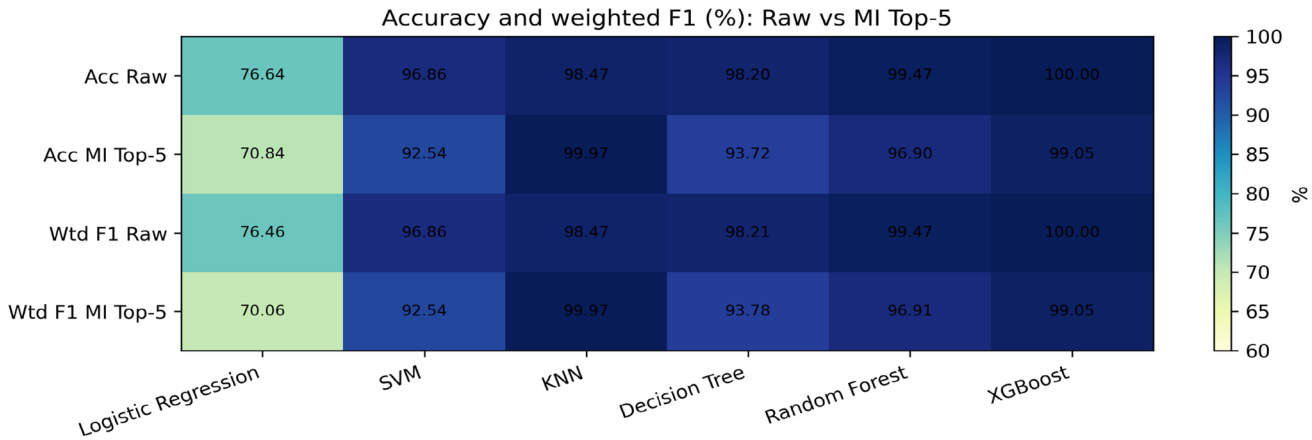


Fig. 14 Heatmap of test accuracy and weighted F1-score with all features and with the MI Top-5 features

Table 11 Test-set performance on DDoS-SDN with the MI Top-12 features (%, attack class; Acc relative to all features, percentage points)

Model	Accuracy	Precision	Recall	Specificity	F1-score	ROC-AUC	Acc
Logistic regression	74.39	69.53	61.14	82.87	65.07	0.8169	−2.25
SVM	95.88	94.72	94.73	96.62	94.73	0.9943	−0.97
KNN	99.31	99.21	99.02	99.49	99.11	0.9990	+0.84
Decision tree	98.23	97.44	98.05	98.35	97.74	0.9922	+0.03
Random forest	99.40	98.94	99.53	99.32	99.23	0.9999	−0.07
XGBoost	99.99	100.00	99.98	100.00	99.99	1.0000	−0.01

ability, because the 151,200 records reduce to 65 distinct feature vectors that all occur in both the training and the test portions (Table 8). E1-Top-5, the five highest-ranked E1 features (frame.len, ip.flags.df, bytes, tx bytes and rx bytes), does not include the PSH flag, and E2 removes the PSH flag from E1. In both settings every classifier reaches between 75.35% and 75.82% accuracy (Fig. 25), at the ceiling of about 75.8% imposed by the 21 feature vectors that occur with more than one label (75.78% over the whole dataset; small differences reflect the composition of the test portion). Benign traffic is still recognised, but the two attack classes cannot be told apart. E3 uses the E2 features for binary benign-versus-attack detection; every classifier reaches 100%, and frame.len, ip.flags.df or bytes alone also separates the two classes perfectly.

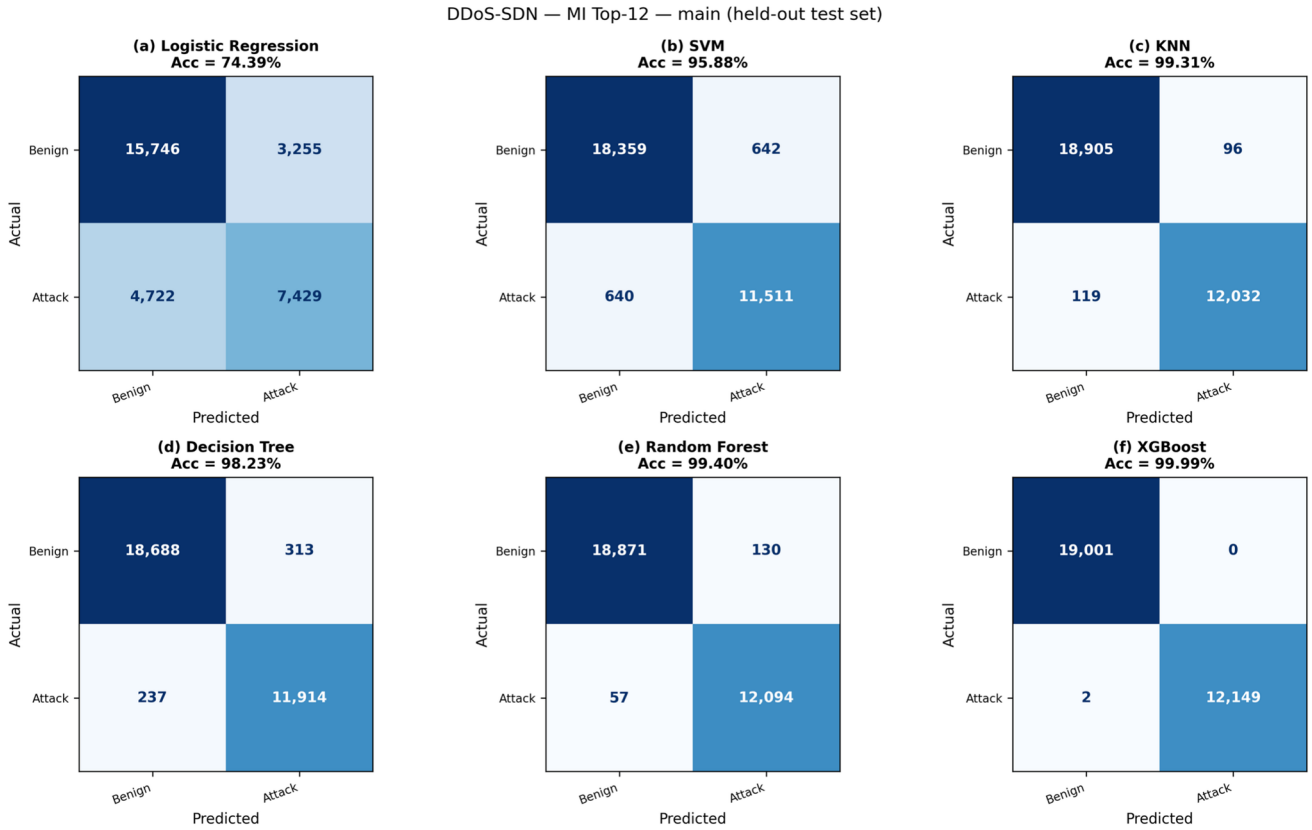


Fig. 15 Confusion matrices of the six classifiers on the DDoS-SDN test set with the MI Top-12 features

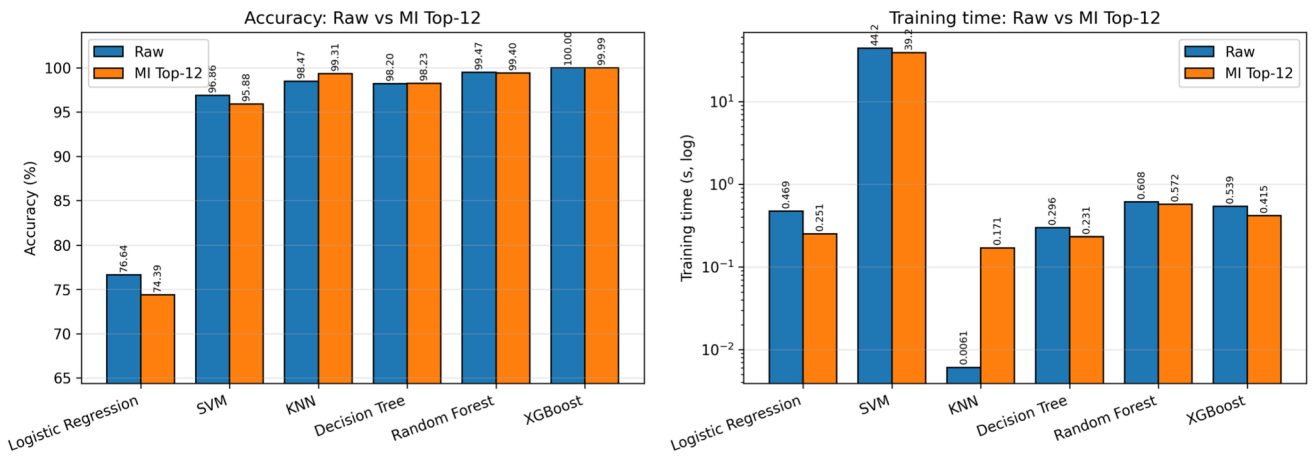


Fig. 16 Test accuracy and training time with all 19 features and with the MI top-12 features

Figure 26 summarises the test accuracy of each classifier under all configurations.

Table 17 also shows that model capacity does not explain the APA-DDoS results. The unconstrained trees give accuracies within 0.5 percentage points of the constrained models in every configuration: 75.00% for the Decision Tree, 76.07% for the Random Forest and 77.57% for XGBoost on Raw-22, 100% on E1, and 75.80% on E1-Top-5 and E2. Random Forest is also the only model whose cross-validated accuracy varies noticeably between folds on Raw-22 (82.85 ± 7.31%), because whether its trees split on frame.time or on the PSH flag depends on the bootstrap sample. For these reasons, APA-DDoS is not used to rank classifiers in the remainder of the paper.

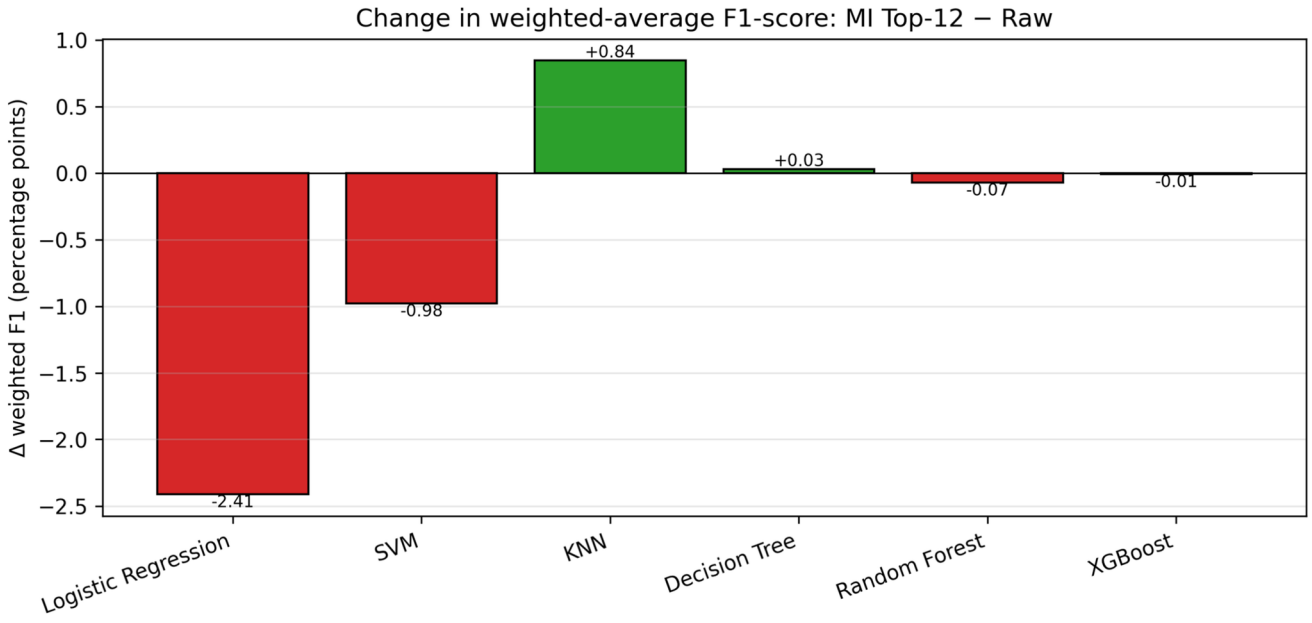


Fig. 17 Change in weighted F1-score after MI Top-12 feature selection

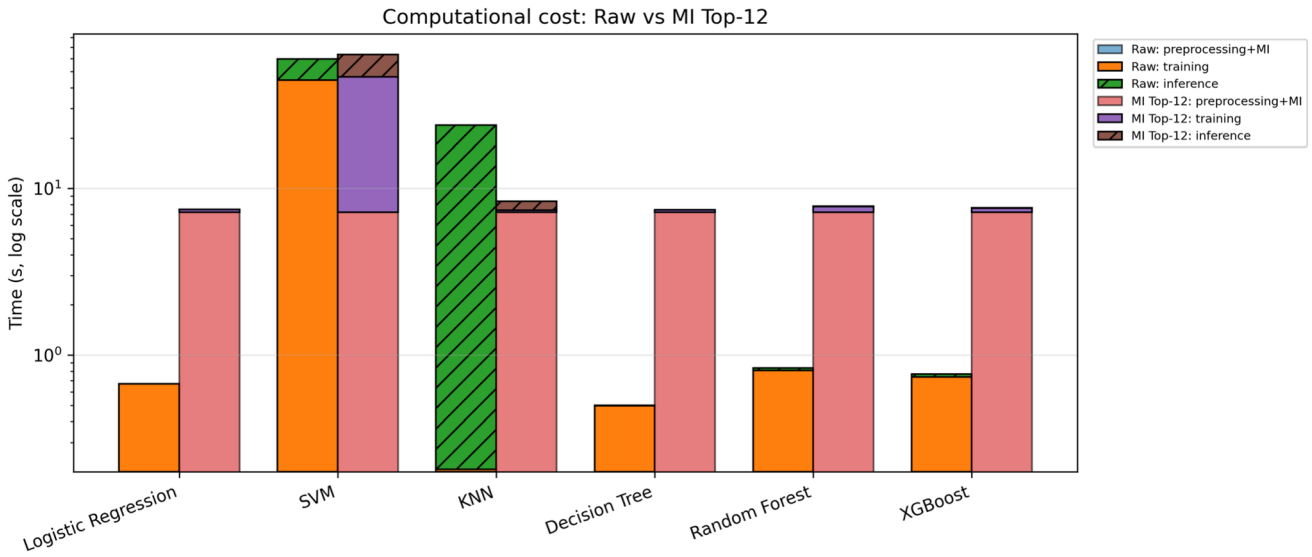


Fig. 18 Computational cost (preprocessing and MI fitting, training and inference) with all 19 features and with the MI Top-12 features

5 Statistical testing and computational analysis

This section complements the per-model performance tables with a statistical, computational and cross-dataset analysis of the results. Section 5.1 applies McNemar's test with Holm correction to determine whether the differences between classifiers on DDoS-SDN are statistically significant on the same test records, and Sect. 5.2 reports their computational cost in terms of training time, inference latency and throughput. Section 5.3 then checks whether the constrained hyperparameters limit the tree-based models, and Sect. 5.4 compares the two datasets and explains why the conclusions of the paper are drawn from DDoS-SDN.

5.1 McNemar's test

McNemar's test compares two classifiers on the same test records using only the discordant pairs: b is the number of records that the first model classifies correctly and the second incorrectly, and c is the reverse. The statistic (|b

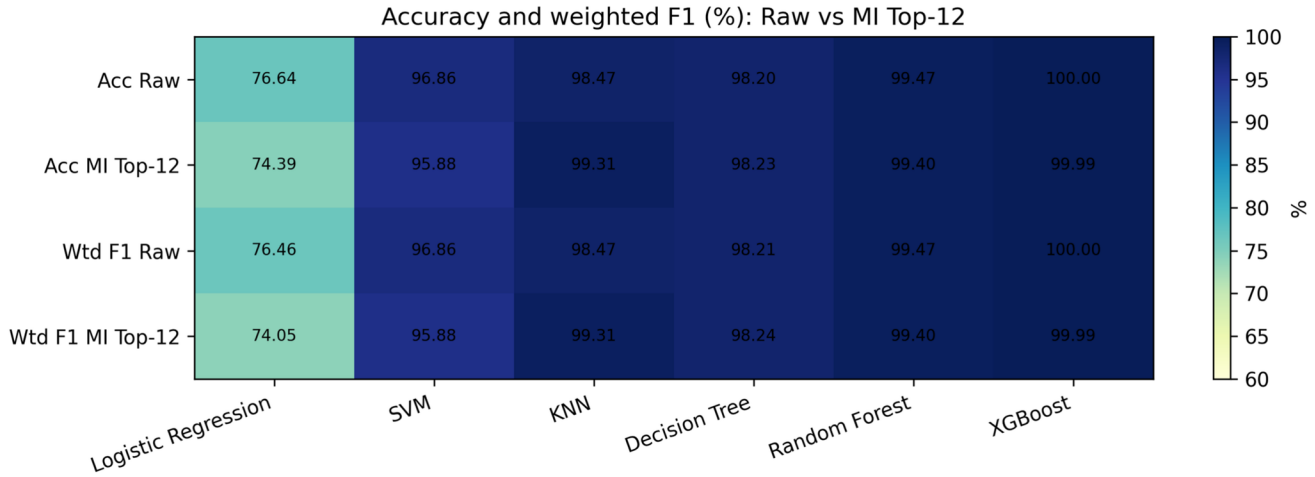


Fig. 19 Heatmap of test accuracy and weighted F1-score with all 19 features and with the MI Top-12 Features

Table 12 Stratified ten-fold cross-validation on the DDoS-SDN training portion, all 19 features (%)

Model	Accuracy (mean ± SD)	Weighted precision	Weighted recall	Weighted F1 (mean ± SD)	ROC-AUC (%)
Logistic regression	76.93 ± 0.54	76.69	76.93	76.72 ± 0.56	84.14
KNN	98.26 ± 0.22	98.26	98.26	98.26 ± 0.22	99.67
Decision tree	98.14 ± 0.19	98.17	98.14	98.14 ± 0.18	99.30
Random forest	99.35 ± 0.11	99.35	99.35	99.35 ± 0.11	99.99
XGBoost	100.00 ± 0.00	100.00	100.00	100.00 ± 0.00	100.00

Table 13 Stratified ten-fold cross-validation on the DDoS-SDN training Portion, MI Top-5 features (%)

Model	Accuracy (mean ± SD)	Weighted precision	Weighted recall	Weighted F1 (mean ± SD)	ROC-AUC (%)
Logistic regression	71.25 ± 0.54	70.71	71.25	70.45 ± 0.58	77.00
KNN	99.93 ± 0.04	99.93	99.93	99.93 ± 0.04	99.97
decision tree	94.19 ± 0.28	94.56	94.19	94.23 ± 0.27	98.03
Random forest	96.66 ± 0.23	96.71	96.66	96.67 ± 0.23	99.67
XGBoost	99.09 ± 0.10	99.09	99.09	99.09 ± 0.10	99.96

Table 14 Stratified ten-fold cross-validation on the DDoS-SDN training portion, MI Top-12 features (%)

Model	Accuracy (mean ± SD)	Weighted precision	Weighted recall	Weighted F1 (mean ± SD)	ROC-AUC (%)
Logistic regression	74.68 ± 0.61	74.33	74.68	74.29 ± 0.62	82.24
KNN	99.23 ± 0.13	99.23	99.23	99.23 ± 0.13	99.87
Decision tree	98.20 ± 0.10	98.21	98.20	98.20 ± 0.10	99.08
Random forest	99.42 ± 0.10	99.42	99.42	99.42 ± 0.10	99.99
XGBoost	100.00 ± 0.00	100.00	100.00	100.00 ± 0.00	100.00

$-\,c|-1)^2 / (b + c)$ follows a χ^2 distribution with one degree of freedom; when $b + c < 25$, an exact binomial test is used instead. Because 15 pairs are compared for each feature set, the p-values were adjusted with the Holm procedure. Table 18 reports the results with all 19 DDoS-SDN features. All 15 pairs differ significantly after correction; even the

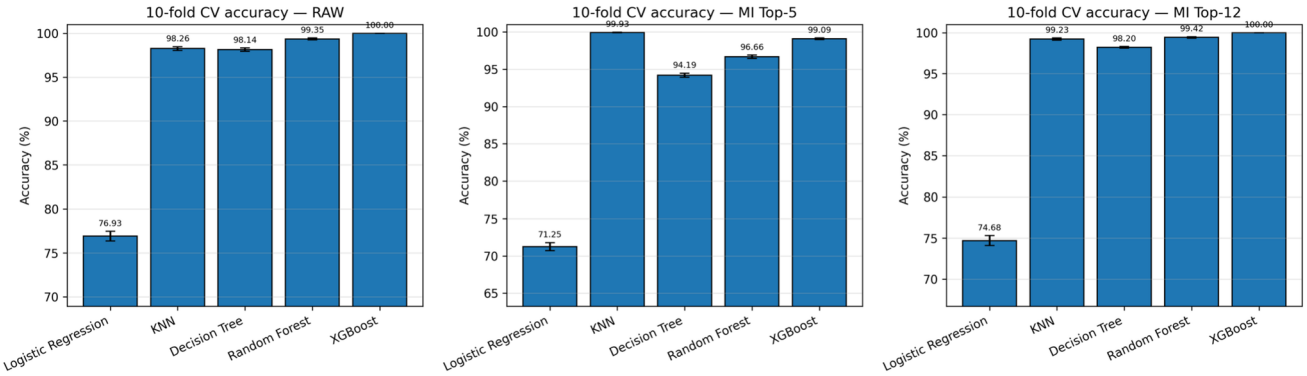


Fig. 20 Ten-fold cross-validation accuracy (mean ± SD) on DDoS-SDN: (left) all 19 features, (centre) MI Top-5, (right) MI Top-12

Table 15 Test accuracy (%) on DDoS-SDN under the random split and three stricter evaluations

Feature set	Model	Random split	Time-ordered	Unseen hosts	Without switch/ port_no
All 19	Logistic regression	76.64	70.35	74.30	76.70
All 19	SVM	96.86	88.69	94.56	96.56
All 19	KNN	98.47	85.83	93.85	99.04
All 19	Decision tree	98.20	80.44	97.59	98.20
All 19	Random forest	99.47	91.11	98.63	99.46
All 19	XGBoost	100.00	93.33	99.33	100.00
MI Top-12	Logistic regression	74.39	72.75	73.22	74.39
MI Top-12	SVM	95.88	86.39	93.82	95.88
MI Top-12	KNN	99.31	85.76	94.45	99.31
MI Top-12	Decision tree	98.23	80.44	97.49	98.23
MI Top-12	Random forest	99.40	89.98	98.18	99.40
MI Top-12	XGBoost	99.99	79.86	99.29	99.99

closest pair, KNN and the Decision Tree, has 466 and 384 discordant records (adjusted $p = 0.0055$). For Random Forest and XGBoost, XGBoost is correct on all 165 records on which the two models disagree. With the Top-5 subset all 15 pairs differ, and with the Top-12 subset all pairs differ except KNN and Random Forest ($b = 166$, $c = 194$, adjusted $p = 0.155$). A significant result indicates that two classifiers make different errors on the same records; with more than 31,000 test records even small differences are detected, so the practical size of a difference should be judged from Tables 9, 10, and 11. On APA-DDoS with the Top-5 subset, five classifiers produce identical predictions, so the corresponding pairs have no discordant records.

5.2 Computational cost

Table 19 reports the computational cost on DDoS-SDN, measured on a CPU only (Intel Xeon 2.20 GHz, four logical cores, 31.3 GB RAM). Training times are means over three repetitions (one for SVM), and inference times are means over ten predictions of the full test set of 31,152 records; latency is the inference time per record and throughput its inverse. Fitting the preprocessing and MI selection takes 7.2 s for the feature-selected sets, compared with 0.2 s for imputation, encoding and scaling alone, but this is a one-off cost paid at training time. At inference time, the tree-based models and Logistic Regression classify a record in less than 0.001 ms regardless of the feature set, whereas the instance-based KNN and the kernel SVM are the most expensive models. Feature selection has the largest effect on KNN, whose latency falls from 0.760 ms to 0.007 ms per record with the Top-5 subset (throughput rising from about 1,300 to 151,000 records per second) and to 0.031 ms with the Top-12 subset. SVM remains the slowest model to train (39–46 s) and to apply (0.49–0.66 ms per record) for every feature set. Figures 27 and 28 show the latency and throughput of all models. These are offline measurements; they indicate relative cost and feasibility rather than performance inside a running SDN controller.

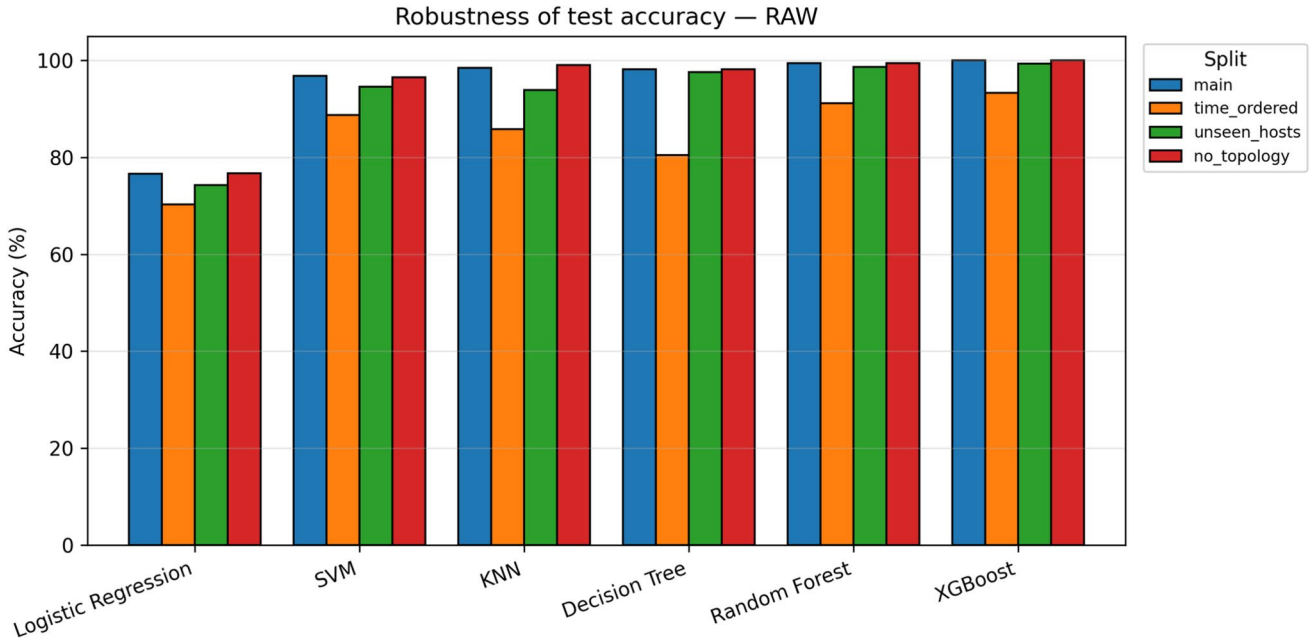


Fig. 21 Test accuracy on DDoS-SDN (all 19 features) under the random split, time-ordered split, unseen-host split and without topology attributes

Table 16 APA-DDoS test-set performance (%) with the original 22 attributes and the MI Top-5 and Top-12 subsets (macro-averaged F1-score; macro one-vs-rest ROC-AUC)

Model	Raw-22 Acc	Raw-22 F1	Top-5 Acc	Top-5 F1	Top-12 Acc	Top-12 F1	Raw-22 ROC-AUC
Logistic regression	100.00	100.00	52.72	40.91	100.00	100.00	1.0000
SVM	99.97	99.96	75.00	55.56	99.97	99.96	1.0000
KNN	99.73	99.64	75.00	55.56	99.73	99.64	0.9976
Decision tree	75.00	55.56	75.00	55.56	75.00	55.56	0.7778
Random forest	76.57	59.96	75.00	55.56	95.07	93.37	1.0000
XGBoost	77.57	62.57	75.00	55.56	77.57	62.57	1.0000

5.3 Model capacity

Table 20 tests whether the constrained hyperparameters of Sect. 3.2 limit the tree-based models on DDoS-SDN by comparing them with an unconstrained Decision Tree, a Random Forest of 200 fully grown trees and an XGBoost model with 500 trees of depth 10, trained and tested on the same split. Removing the constraints raises the accuracy of the Decision Tree by 1.76 percentage points with all features and by 6.24 with the Top-5 subset, and that of the Random Forest by 0.53 and 3.08, so that both reach at least 99.96%. XGBoost changes by at most 0.34 percentage points. The constrained settings therefore trade a small amount of accuracy for smaller models on DDoS-SDN, and the lower Top-5 accuracy of the constrained trees is mainly a capacity effect rather than a loss of information. On APA-DDoS, by contrast, the unconstrained models give the same accuracies as the constrained ones (Table 17), which shows directly that the APA-DDoS results are determined by the features rather than by model capacity.

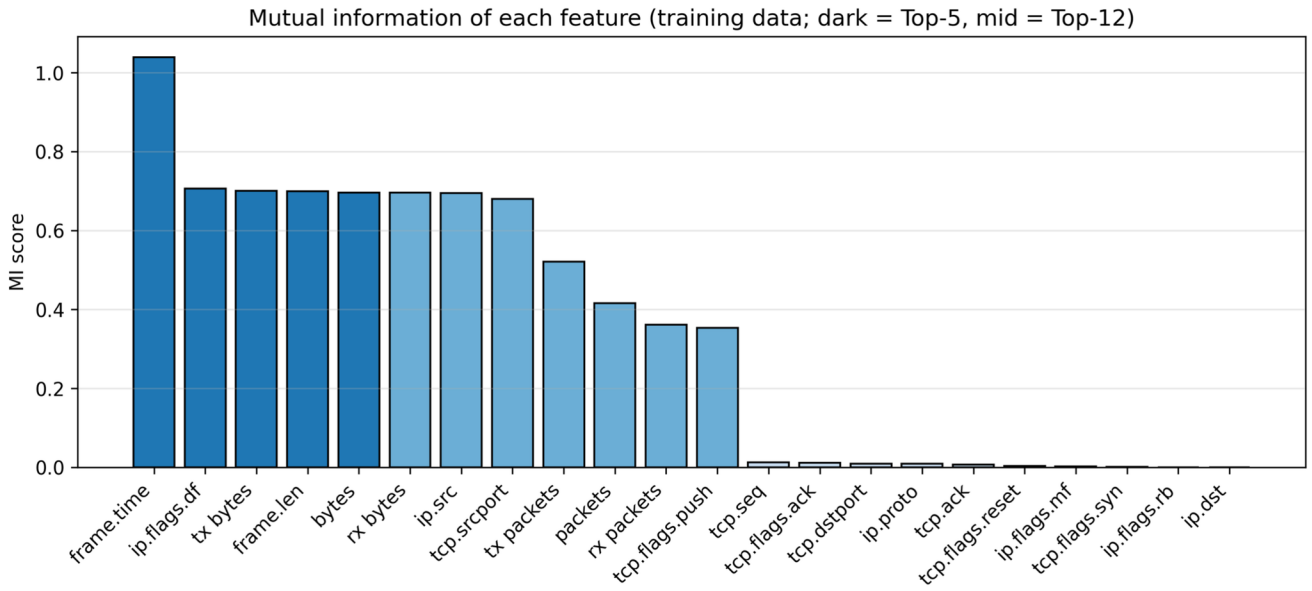


Fig. 22 Mutual information score of each of the 22 APA-DDoS attributes (training portion; dark = Top-5, Mid = Top-12)

Table 17 APA-DDoS test accuracy (%) under the audit configurations (E0 = raw-22; E1 = identifiers and constants removed; E1-Top-5 = MI Top-5 of E1; E2 = E1 without tcp. flags.push; E3 = E2 features, benign vs attack)

Model (setting)	E0	E1	E1-Top-5	E2	E2 macro F1	E3
Logistic regression (constrained)	100.00	100.00	75.70	75.70	67.21	100.00
SVM (constrained)	99.97	100.00	75.82	75.80	63.82	100.00
KNN (constrained)	99.73	100.00	75.57	75.35	66.78	100.00
Decision tree (constrained)	75.00	100.00	75.57	75.57	66.51	100.00
Random forest (constrained)	76.57	100.00	75.81	75.81	63.83	100.00
XGBoost (constrained)	77.57	100.00	75.80	75.80	63.81	100.00
Decision tree (unconstrained)	75.00	100.00	75.80	75.80	63.81	100.00
Random forest (unconstrained)	76.07	100.00	75.80	75.80	63.81	100.00
XGBoost (unconstrained)	77.57	100.00	75.80	75.80	63.81	100.00

5.4 Comparison across the two datasets

The two datasets lead to different conclusions, and the reason lies in the data rather than in the classifiers. On DDoS-SDN, the 19 behavioural flow features support a clear and stable ranking (XGBoost, Random Forest, KNN, Decision Tree, SVM and Logistic Regression), which is confirmed by cross-validation and McNemar's test and is unaffected by the removal of the topology attributes. On APA-DDoS, the apparent ranking reverses depending on how the timestamp is handled, every classifier is perfect once identifiers are removed, and every classifier is limited to about 75.8% once the PSH flag is also removed. APA-DDoS is therefore suitable for illustrating dataset leakage and redundancy but not for comparing classifiers, and the conclusions of this paper about classifier performance and feature selection are drawn from DDoS-SDN only.

6 Discussion

The feature selection approach employed in [22] uses Vaccine-Based Binary Grey Wolf Optimization (Vaccine-BGWO), a wrapper method that optimises the feature subset with a fitness function balancing classification error against the number of selected features; it reported 99.99% and 100% accuracy for Random Forest and XGBoost after optimisation, compared with 99.99% and 99.80% on the raw features. The present study uses Mutual Information as a filter method, which is cheaper and independent of the

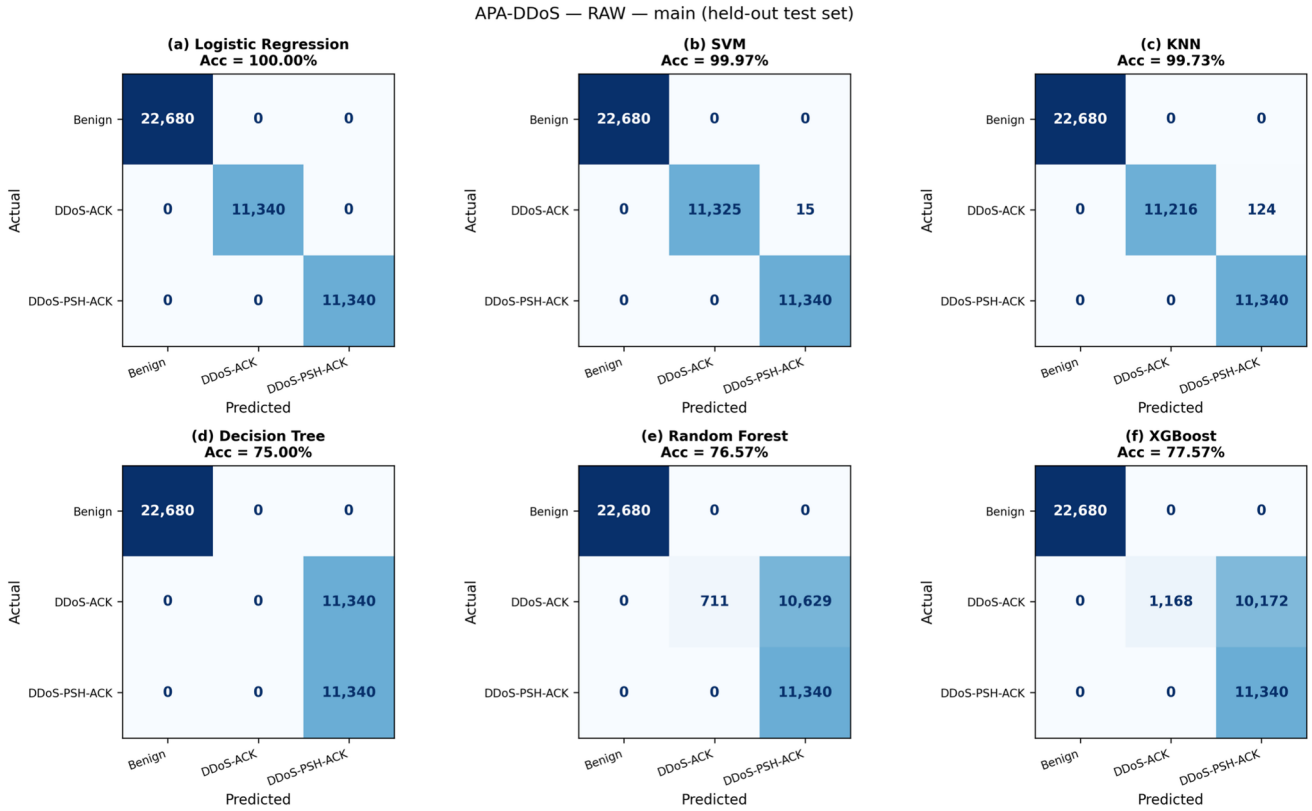


Fig. 23 Confusion matrices of the six classifiers on the APA-DDoS test set with the original 22 attributes (Raw-22)

classifier, and obtains comparable results on DDoS-SDN: Random Forest and XGBoost reach 99.47% and 100% with all 19 features and 99.40% and 99.99% with the Top-12 subset. Three further findings qualify these numbers. First, near-perfect accuracy under a random split is not the same as generalisation: with a time-ordered split, accuracy falls by 6.7–17.8 percentage points for the non-linear classifiers, and 6.55% of the random-split test records have an identical feature vector in the training data, so random-split accuracies are an upper bound for traffic recorded under the same conditions. Second, the effect of feature selection depends on the classifier: it improves KNN in both accuracy and latency and costs little for the ensembles with twelve features, while its larger cost for the constrained trees with five features largely disappears when the trees are unconstrained. Third, the APA-DDoS audit shows that a benchmark can reward models for exploiting identifiers and label-defining attributes, which is why a feature-level audit should precede any comparison of classifiers.

6.1 Limitations of the study

Several limitations should be noted. First, the comparison of classifiers rests on a single dataset, DDoS-SDN, generated in an emulated Mininet environment; the results may not transfer directly to production SDN traffic or to previously unseen attack types, and the time-ordered evaluation shows that performance on later traffic is lower than under a random split. Second, feature selection relies on a single filter method, Mutual Information, with two subset sizes fixed in advance; wrapper, embedded and metaheuristic approaches such as Vaccine-BGWO were not compared and might select different subsets. Third, APA-DDoS could not be used to rank classifiers because of identifier leakage, the construction-level definition of its attack classes and the small number of distinct feature vectors, so its results are reported as a dataset audit; the E0b result, in which frame.time is treated as a continuous time value, is taken from an earlier run of the same pipeline. Fourth, the hyperparameters were fixed a priori rather than tuned, and SVM was excluded from cross-validation because of its computational cost, so its results rest on the held-out test set alone. Finally, the study addresses detection only and does not implement mitigation inside a live SDN controller; the reported latency and throughput are offline CPU measurements rather than measurements under real-time load, and deep learning architectures such as CNN-LSTM and Transformer models were outside the

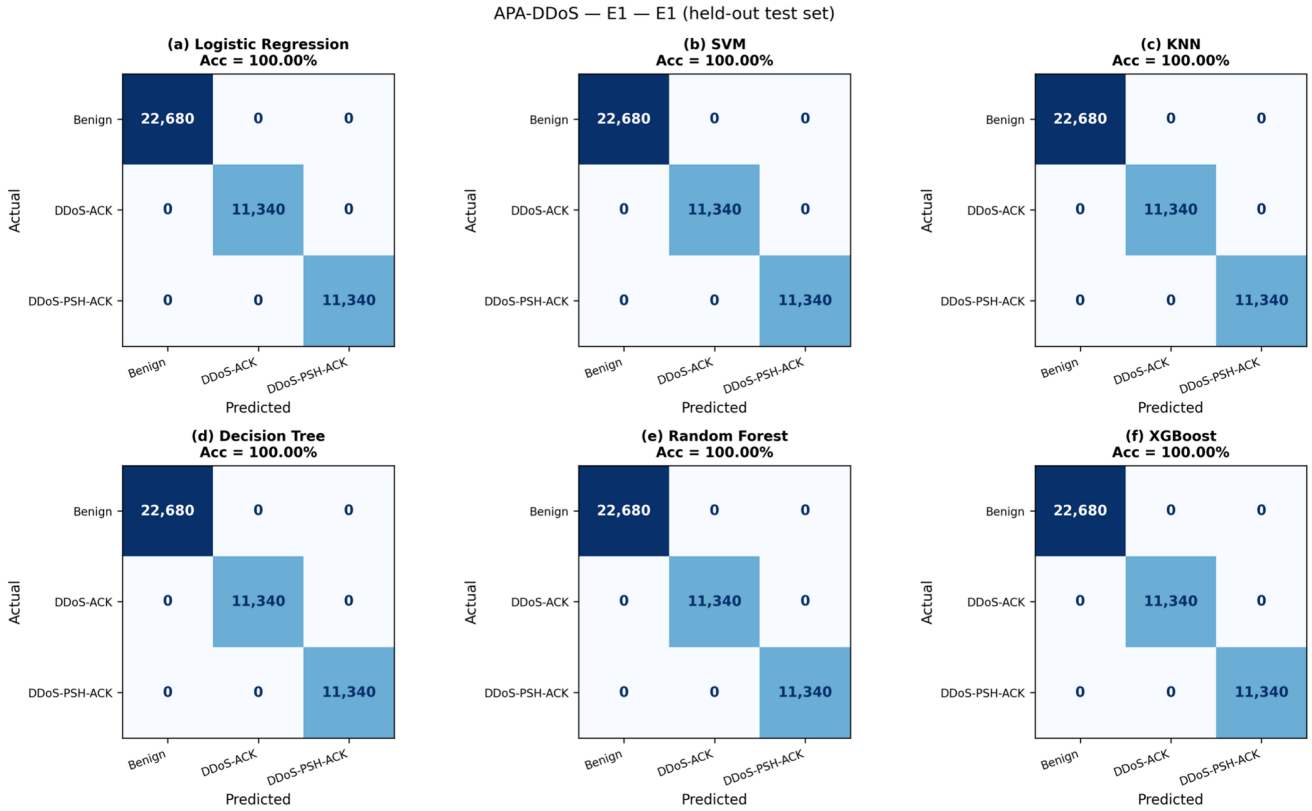


Fig. 24 Confusion matrices on the APA-DDoS test set after removal of identifier and constant attributes (E1, nine features)

scope of this comparison. These limitations motivate the future directions outlined below.

7 Conclusion and future work

This paper compared six supervised classifiers for DDoS detection in SDN under a leakage-controlled protocol in which every preprocessing and feature-selection step is fitted on the training data only. On the DDoS-SDN dataset, XGBoost and Random Forest were the strongest classifiers, reaching 100% and 99.47% test accuracy with all 19 features, and this ranking was confirmed by ten-fold cross-validation and McNemar's test with Holm correction. Mutual Information feature selection reduced the input dimensionality with little loss: the Top-12 subset retained 99.99% accuracy for XGBoost and 99.40% for Random Forest, and the Top-5 subset raised KNN accuracy to 99.97% while reducing its inference latency by two orders of magnitude. A time-ordered split reduced accuracy by 6.7–17.8 percentage points, showing that random-split results overstate performance on later traffic. On the APA-DDoS dataset, a feature-level audit showed that the capture timestamp encodes the class label, that the TCP PSH flag defines the two attack classes, and that the records reduce to 65 distinct feature vectors once identifiers are removed; the dataset therefore cannot discriminate between classifiers, which underlines the need to audit input features before comparing models. Future work will evaluate the pipeline on traffic captured in a live SDN controller, compare wrapper and embedded feature-selection methods, and explore temporal

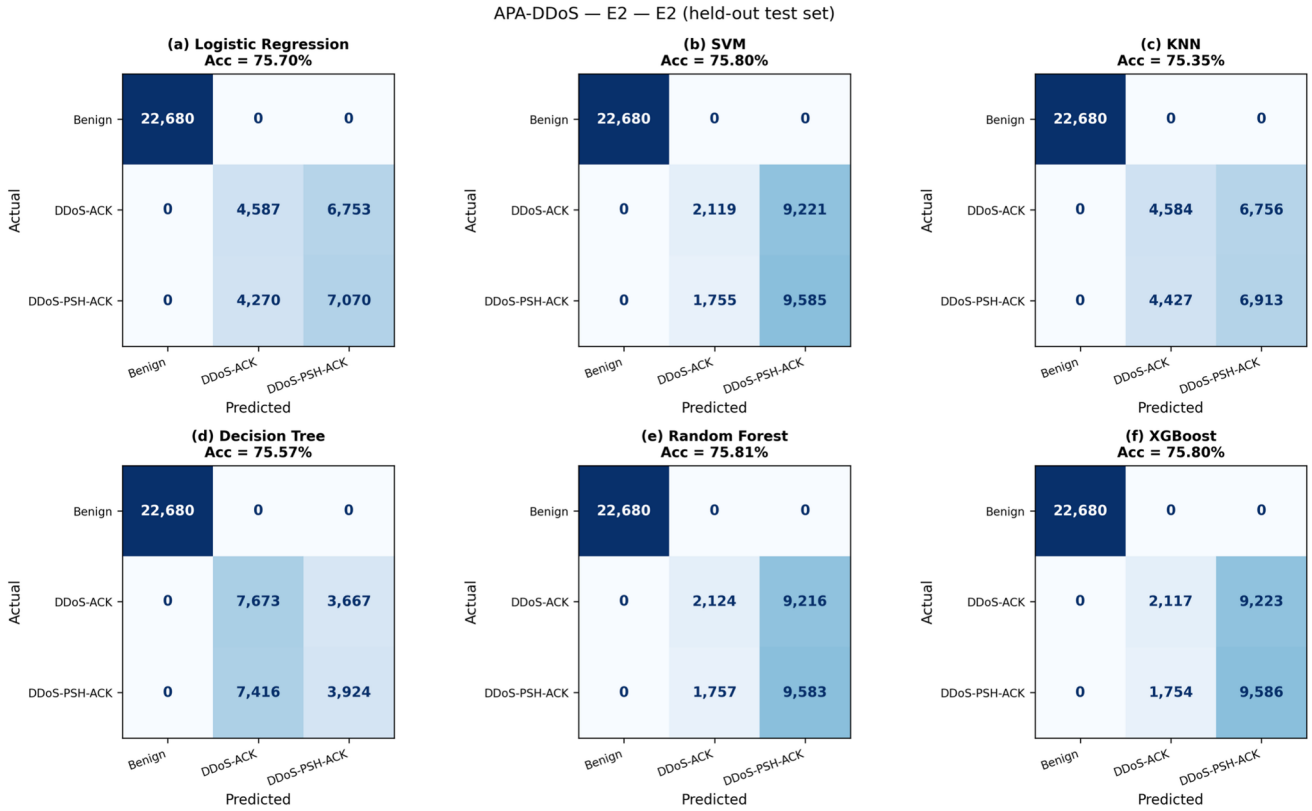


Fig. 25 Confusion matrices on the APA-DDoS test set without the TCP PSH flag (E2, eight features)

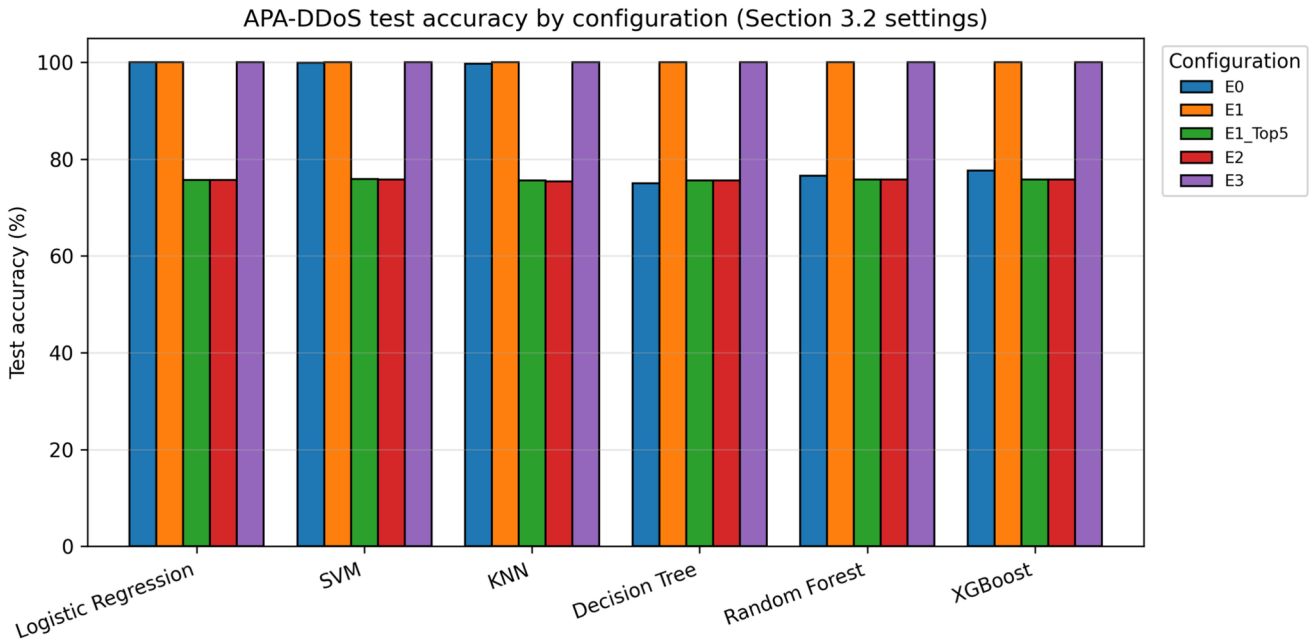


Fig. 26 APA-DDoS test accuracy of each classifier under the audit configurations E0, E1, E1-Top-5, E2 and E3

Table 18 McNemar's test between all pairs of classifiers on the DDoS-SDN test set with all 19 features (Holm-adjusted p-Values)

Model 1	Model 2	b	c	χ^2	Adjusted p	Significant
Logistic regression	SVM	363	6,661	5645.25	< 0.001	Yes
Logistic regression	KNN	216	7,015	6390.93	< 0.001	Yes
Logistic regression	Decision tree	250	6,967	6249.78	< 0.001	Yes
Logistic regression	Random forest	73	7,185	6966.98	< 0.001	Yes
Logistic regression	XGBoost	0	7,277	7275.00	< 0.001	Yes
SVM	KNN	158	659	306.00	< 0.001	Yes
SVM	Decision TREE	378	797	148.70	< 0.001	Yes
SVM	Random FOREST	43	857	734.41	< 0.001	Yes
SVM	XGBoost	0	979	977.00	< 0.001	Yes
KNN	Decision tree	466	384	7.72	0.0055	Yes
KNN	Random forest	75	388	210.25	< 0.001	Yes
KNN	XGBoost	0	478	476.00	< 0.001	Yes
Decision tree	Random forest	64	459	296.82	< 0.001	Yes
Decision tree	XGBoost	0	560	558.00	< 0.001	Yes
Random forest	XGBoost	0	165	163.01	< 0.001	Yes

Table 19 Computational cost on DDoS-SDN (CPU only; training over 3 repetitions, 1 for SVM; inference on the 31,152 test records over 10 repetitions; mean ± SD)

Feature set	Model	Training (s)	Inference, test set (s)	Latency (ms/record)	Throughput (records/s)
All 19	Logistic regression	0.469 ± 0.029	0.001 ± 0.001	3.5×10^{-5}	28,887,731
All 19	SVM	44.236	15.158 ± 0.021	0.4866	2,055
All 19	KNN	0.006 ± 0.000	23.663 ± 0.082	0.7596	1,316
All 19	Decision tree	0.296 ± 0.002	0.002 ± 0.001	7.5×10^{-5}	13,250,201
All 19	Random forest	0.608 ± 0.003	0.025 ± 0.000	8.2×10^{-4}	1,222,169
All 19	XGBoost	0.539 ± 0.055	0.030 ± 0.000	9.6×10^{-4}	1,043,260
MI Top-5	Logistic regression	0.069 ± 0.001	0.000 ± 0.000	1.4×10^{-5}	69,711,970
MI Top-5	SVM	46.330	20.663 ± 0.021	0.6633	1,508
MI Top-5	KNN	0.073 ± 0.001	0.206 ± 0.005	0.0066	151,382
MI Top-5	Decision tree	0.103 ± 0.001	0.002 ± 0.000	5.6×10^{-5}	17,809,050
MI Top-5	Random forest	0.378 ± 0.006	0.025 ± 0.001	8.1×10^{-4}	1,226,639
MI Top-5	XGBoost	0.284 ± 0.002	0.029 ± 0.000	9.4×10^{-4}	1,061,595
MI Top-12	Logistic regression	0.251 ± 0.012	0.001 ± 0.000	1.9×10^{-5}	51,517,085
MI Top-12	SVM	39.240	16.771 ± 0.031	0.5384	1,858
MI Top-12	KNN	0.171 ± 0.001	0.978 ± 0.017	0.0314	31,843
MI Top-12	Decision tree	0.231 ± 0.002	0.002 ± 0.000	6.1×10^{-5}	16,357,436
MI Top-12	Random forest	0.572 ± 0.004	0.026 ± 0.001	8.2×10^{-4}	1,220,062
MI Top-12	XGBoost	0.415 ± 0.009	0.029 ± 0.000	9.4×10^{-4}	1,061,121

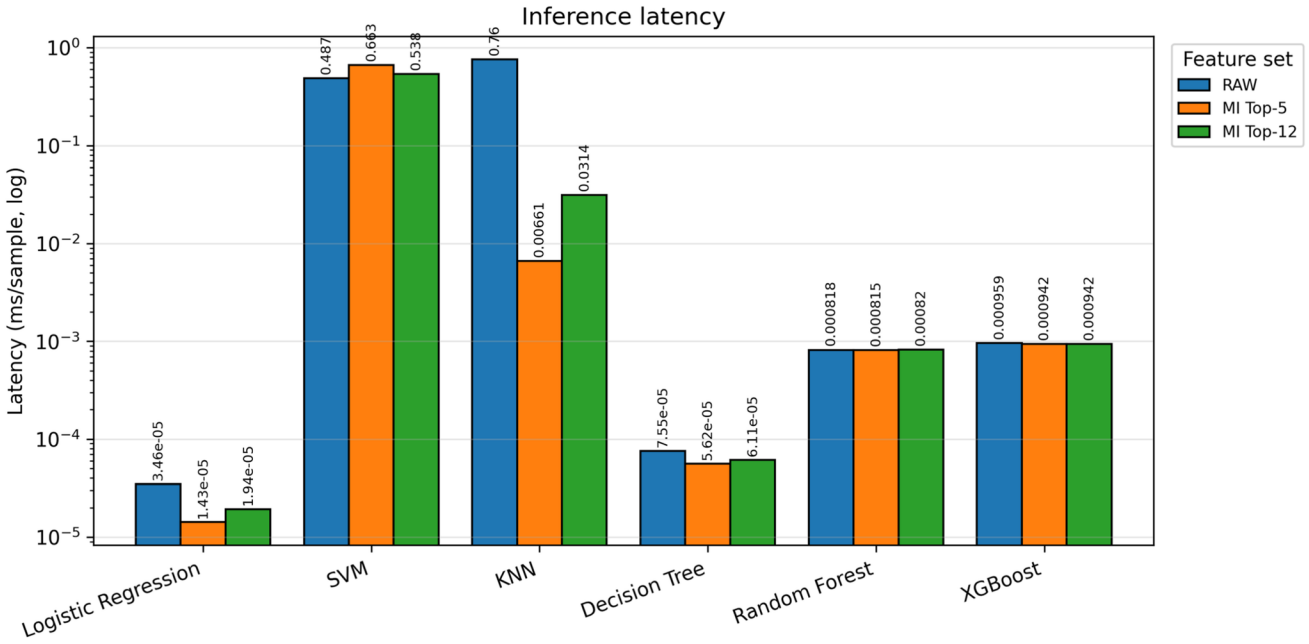


Fig. 27 Inference latency per record on DDoS-SDN for each classifier and feature set (log scale)

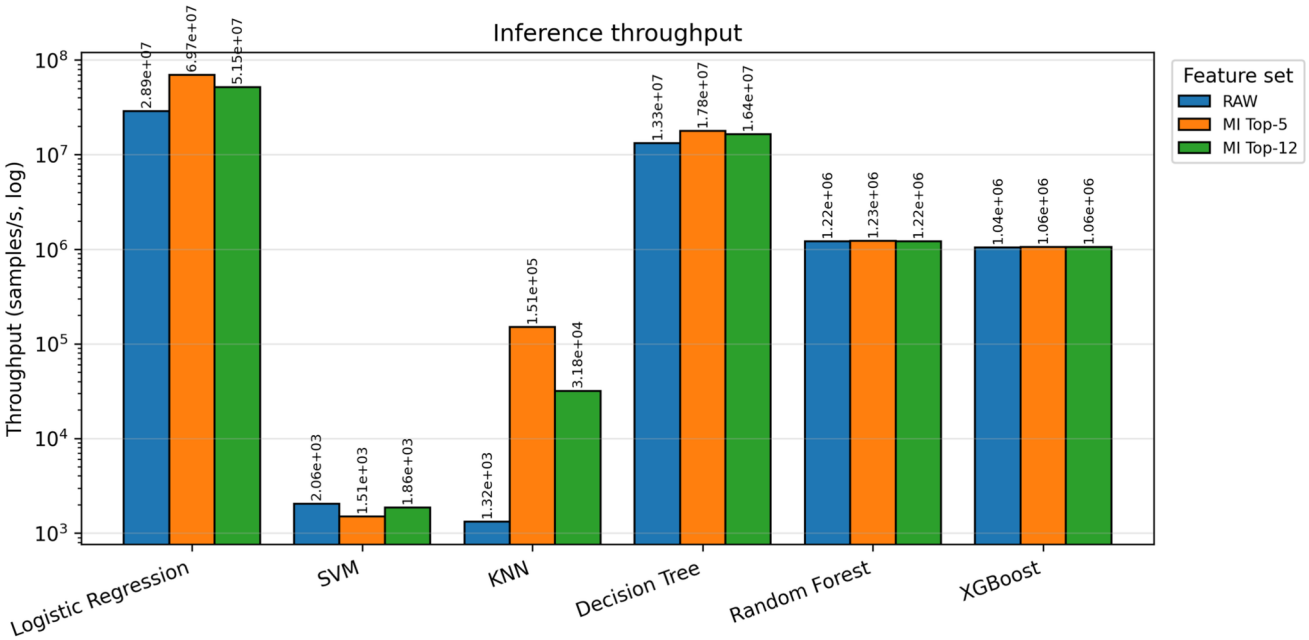


Fig. 28 Inference throughput on DDoS-SDN for each classifier and feature set (log scale)

Table 20 Test accuracy (%) of constrained and unconstrained tree-based models on DDoS-SDN

Feature set	Model	Constrained	Unconstrained	Δ (pp)
All 19	Decision tree	98.20	99.96	+1.76
All 19	Random forest	99.47	100.00	+0.53
All 19	XGBoost	100.00	100.00	+0.00
MI Top-5	Decision tree	93.72	99.96	+6.24
MI Top-5	Random forest	96.90	99.98	+3.08
MI Top-5	XGBoost	99.05	99.39	+0.34
MI Top-12	Decision tree	98.23	99.99	+1.76
MI Top-12	Random forest	99.40	100.00	+0.60
MI Top-12	XGBoost	99.99	99.99	+0.00

deep learning models, online feature selection and explainable detection.

Author contributions A.K.: Conceptualization, Methodology, Software, Investigation, Data curation, Visualization, Writing—original draft. B.S.: Conceptualization, Methodology, Resources, Supervision, Project administration, Writing—review & editing. V.T.: Software, Validation, Investigation, Data curation. P.L.: Methodology, Visualization, Validation, Formal analysis, Writing—review & editing.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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